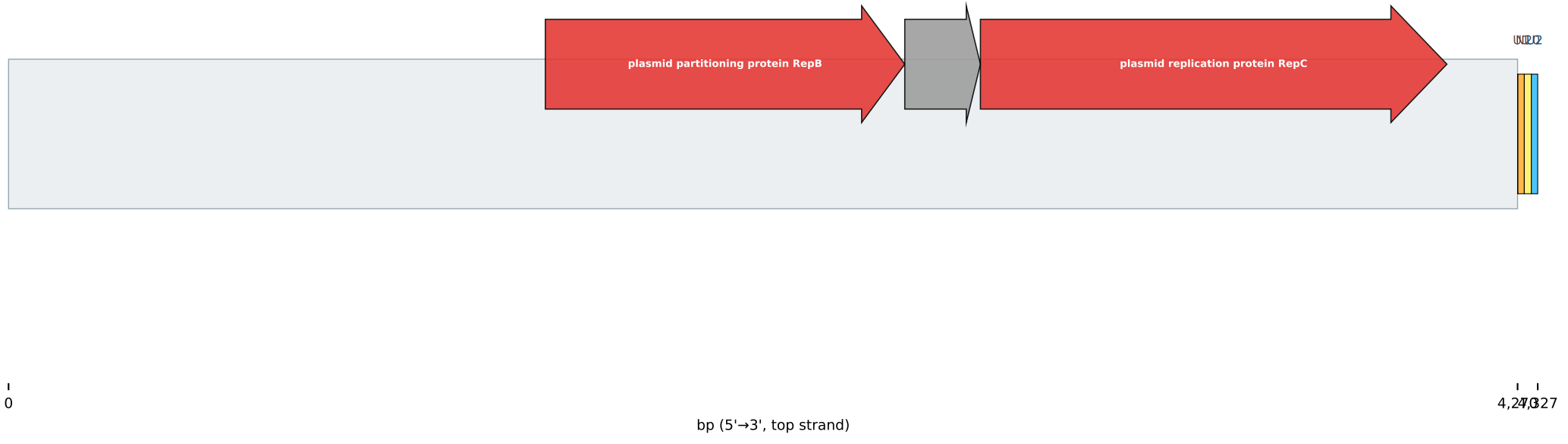
Final Synthesis / Cloning Order

73 origins | 248,974 bp total | 250 kb budget

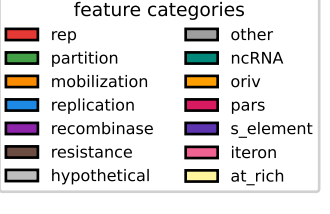
```
Origin bp:                  244,813
Cassette bp (57 bp each):     4,161
Total order bp:             248,974

Pool breakdown:
  repABC_primary               26 origins
  repABC_alt_cloning            7 origins
  diversity_PLSDB              40 origins

Production method:
  twist_synthesis              66 origins
  genomic_pcr_amplification     7 origins

Diversity-tier mix:
  Tier 3: Cross-phylum BHR            5
  Tier 2: Alphaproteobacteria        29
  Tier 4: Phylum-exclusive            6

Barcode cassette (top strand 5'->3'):
  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC    (spacer | U1 | barcode | U2)

Priming sites verified against pKMW7 (Wetmore et al. 2015, mBio).
```

15-1187-1-2a_CP123842.1_pTi_Type_III_rep_1_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 0 misc (.gbk)

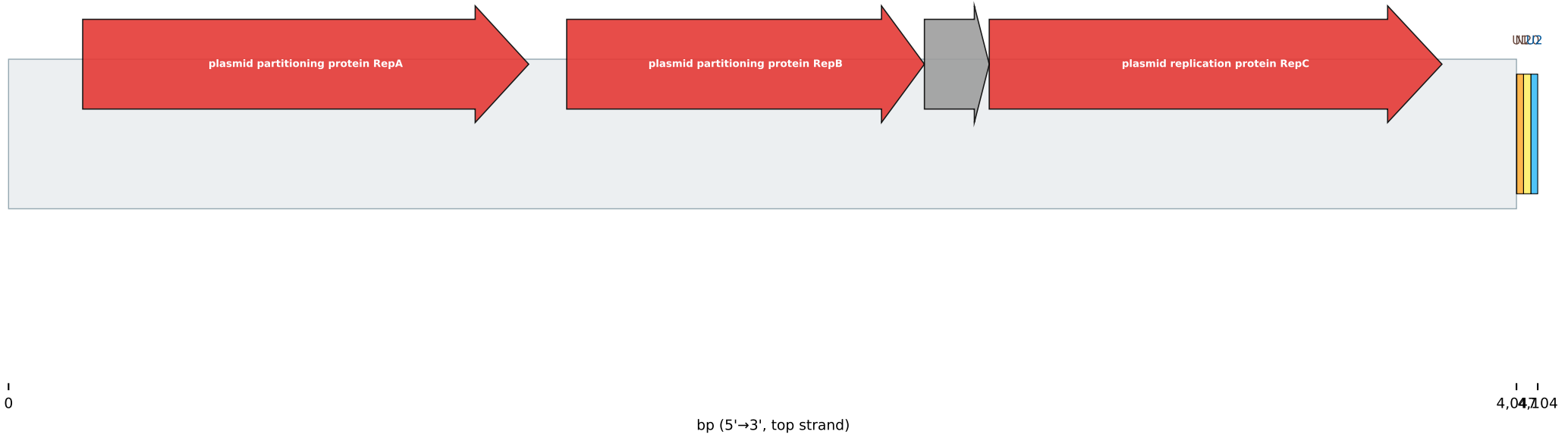



```
Tier / category:     PRIMARY (rank 17)
Rep family:          pTi Type III rep 1
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5478
Barcode (N20):       CGATTCGGTTTGTGAATGAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,327 bp (origin 4,270 + cassette 57)
Flags:               none
```

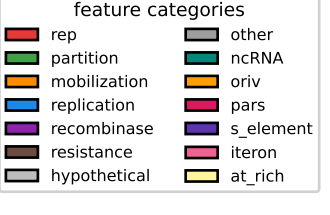


15-1187-1-2a_CP123842.1_pTi_Type_III_rep_2_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 4 CDS + 0 misc (.gbk)

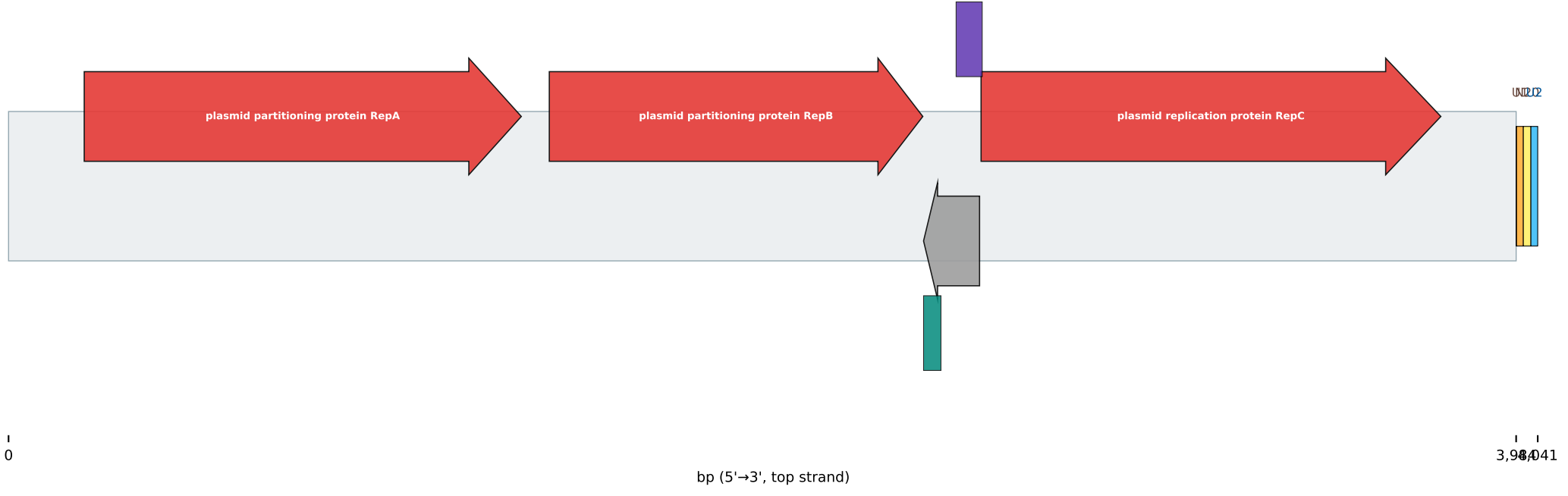



```
Tier / category:     PRIMARY (rank 5)
Rep family:          pTi Type III rep 2
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5382
Barcode (N20):       GATGTTGCGGCAGTTAGCCA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,104 bp (origin 4,047 + cassette 57)
Flags:               none
```



1D1514_plasmid1_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 4 CDS + 2 misc (.gbk)



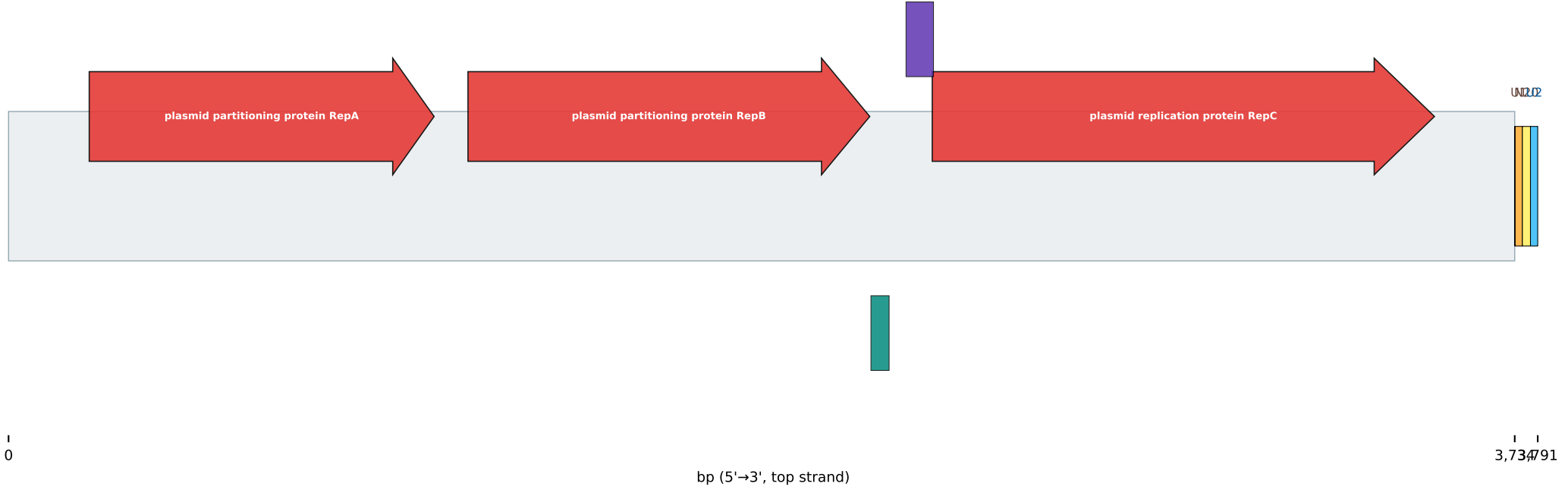

```
Tier / category:     PRIMARY (rank 28)
Rep family:          other_UniRef50_P05684
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5484
Barcode (N20):       CAACGTTATGCTGCCAGCTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,041 bp (origin 3,984 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

1D1514_plasmid2_pTi_Type_V_rep_1_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



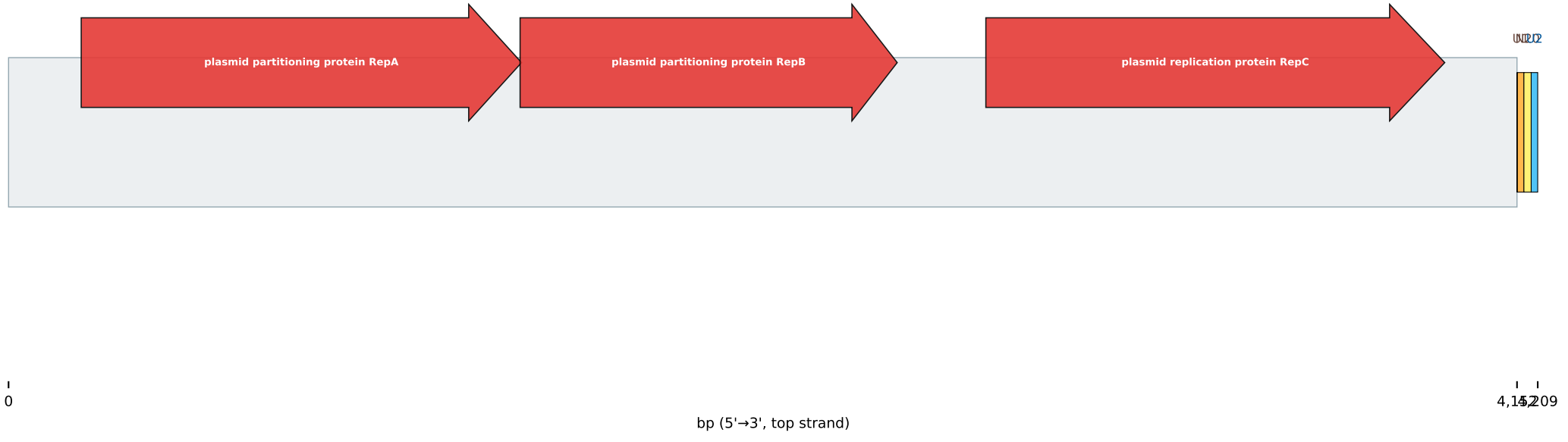

```
Tier / category:     PRIMARY (rank 13)
Rep family:          pTi Type V rep 1
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5463
Barcode (N20):       GTGCGTGATACCACCCTGTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,791 bp (origin 3,734 + cassette 57)
Flags:               none
```

feature categories

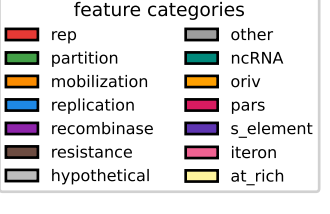
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

1D1514_plasmid2_pTi_Type_V_rep_2_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 0 misc (.gbk)

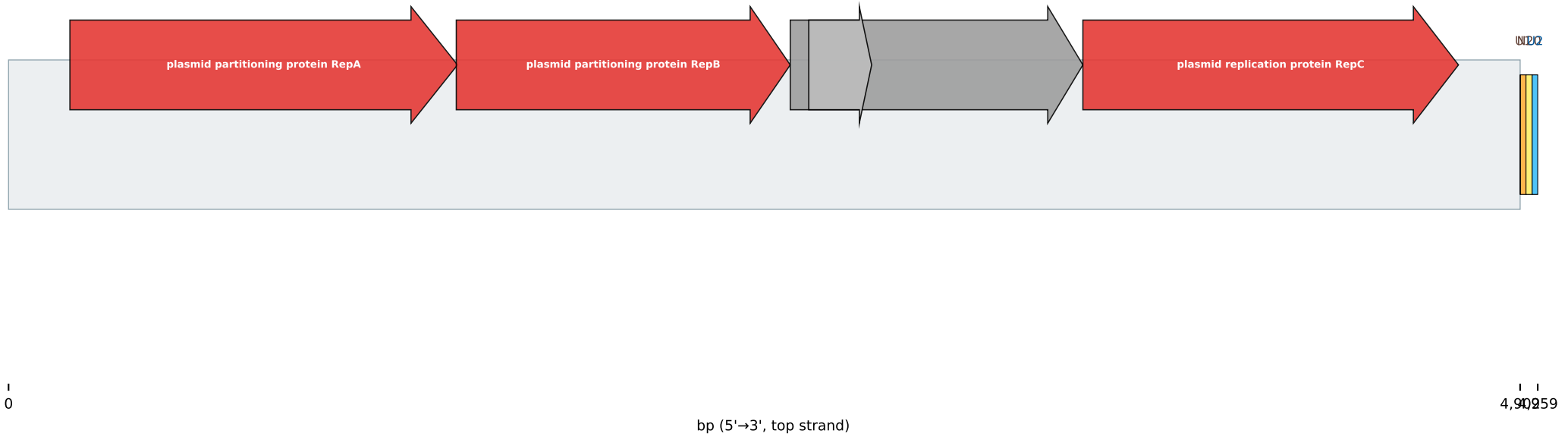



```
Tier / category:     PRIMARY (rank 15)
Rep family:          pTi Type V rep 2
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5226
Barcode (N20):       CAATCGGATGACTCCAGCGA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,209 bp (origin 4,152 + cassette 57)
Flags:               none
```

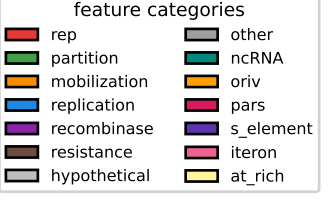


1D1524_chromid_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 5 CDS + 0 misc (.gbk)

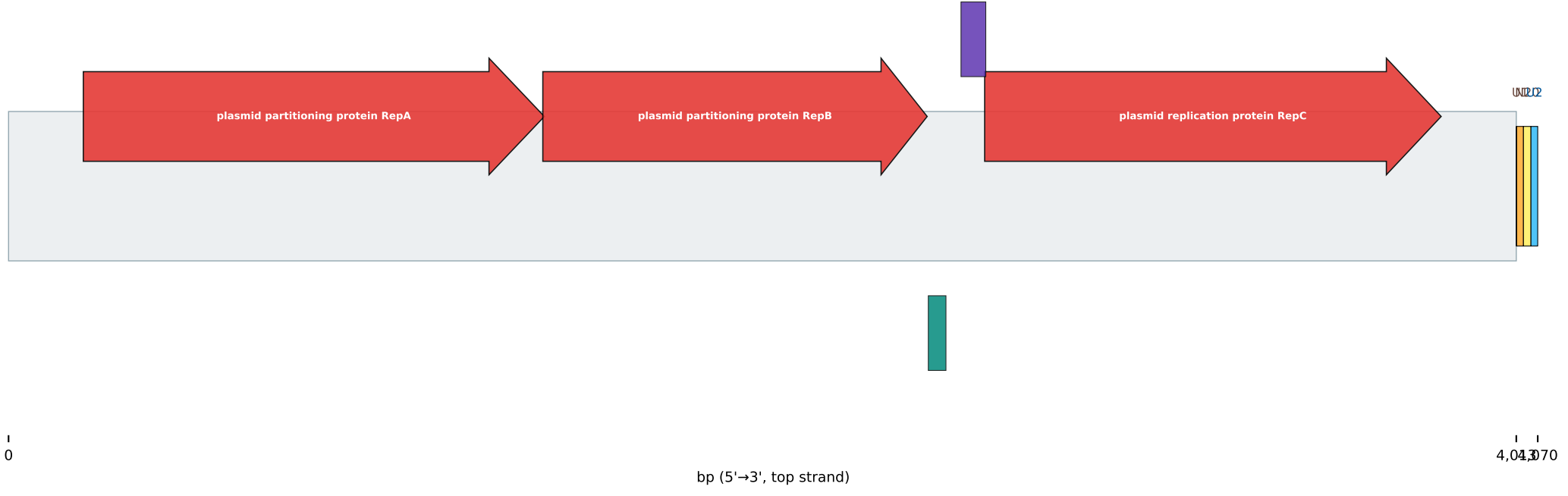



```
Tier / category:     PRIMARY (rank 24)
Rep family:          chromid_UniRef50_A0A6S6NCV8
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5377
Barcode (N20):       ACACCCAAAGCAGCCTTACC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,959 bp (origin 4,902 + cassette 57)
Flags:               none
```



1D1524_pAt_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



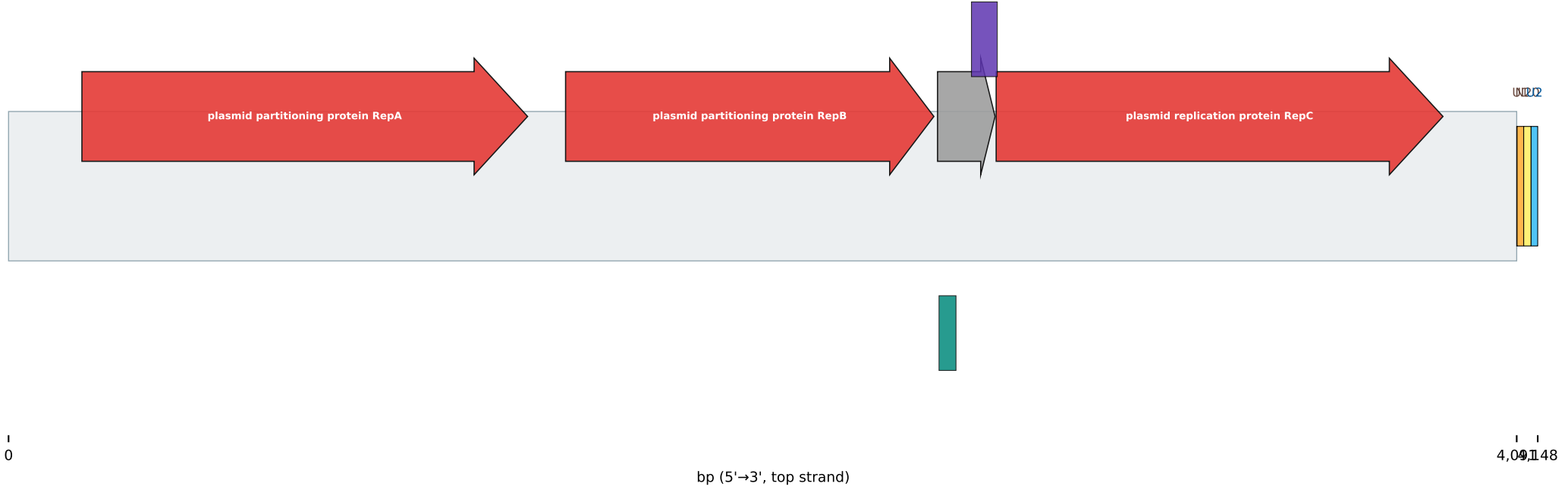

```
Tier / category:     PRIMARY (rank 23)
Rep family:          pAt_UniRef50_P05684
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5136
Barcode (N20):       AATAACAGGCGTTTGGTATC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,070 bp (origin 4,013 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

AF3.44_contig_3_pTi_Type_X_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 4 CDS + 2 misc (.gbk)



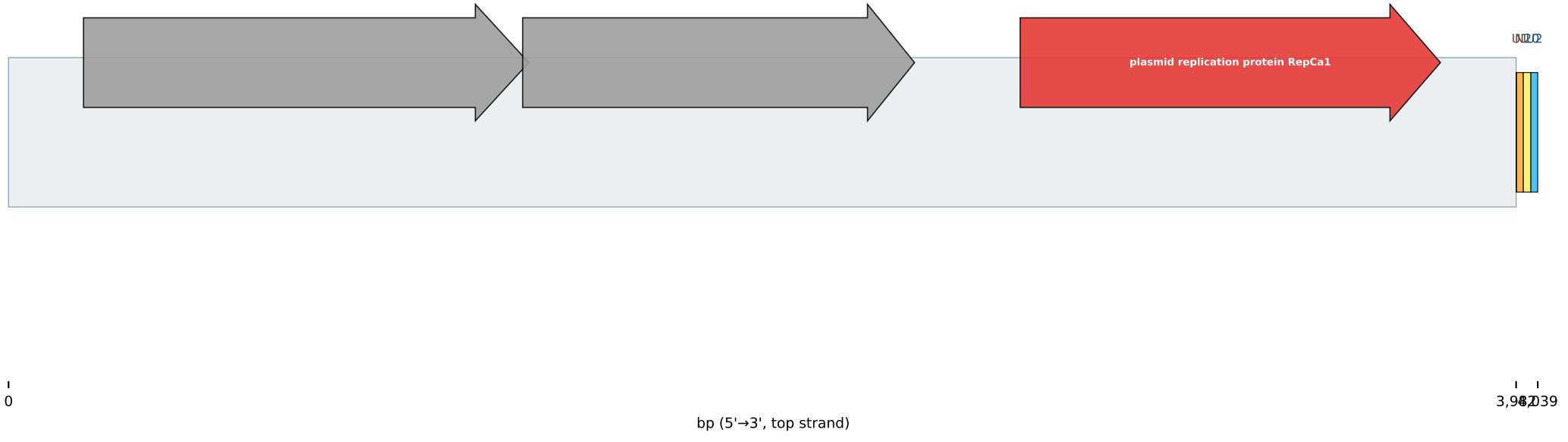

```
Tier / category:     PRIMARY (rank 4)
Rep family:          pTi Type X
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5331
Barcode (N20):       CGGATTGCTAGGTGCCTCAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,148 bp (origin 4,091 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Atu_K84_chromid_CP000629.1_repABC | Agrobacterium radiobacter | repABC_primary | twist_synthesis

annotations: 3 CDS + 0 misc (.gbk)



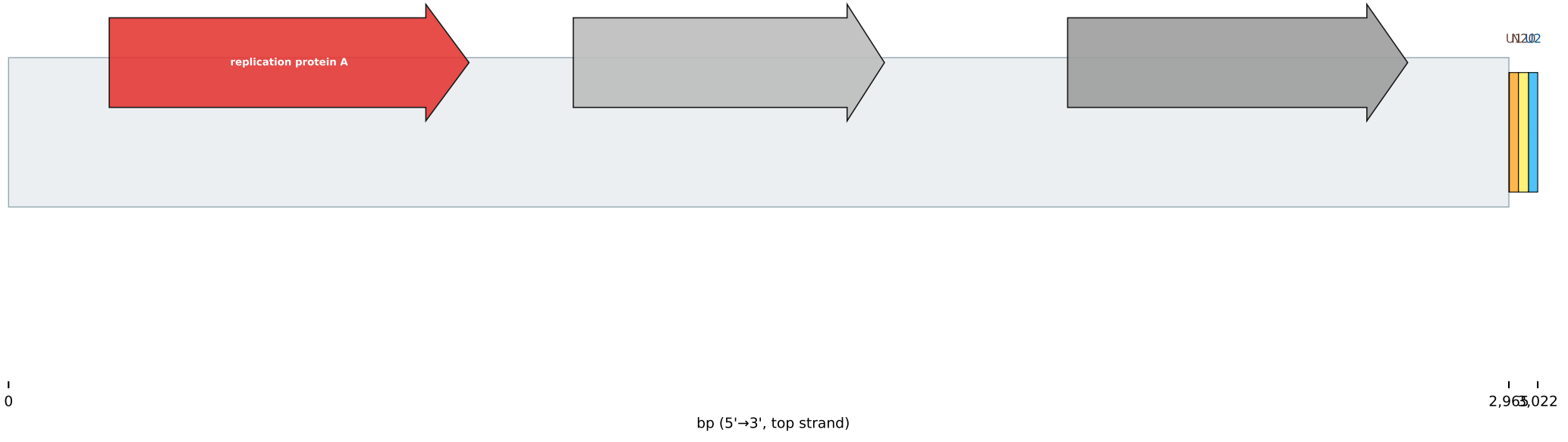

```
Tier / category:     PRIMARY (rank 27)
Rep family:          chromid
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5718
Barcode (N20):       CAGGATTGCCTTGACCACTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,039 bp (origin 3,982 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Atu_K84_pAgK84_CP000632.1_repABC | Agrobacterium radiobacter | repABC_primary | twist_synthesis

annotations: 3 CDS + 0 misc (.gbk)



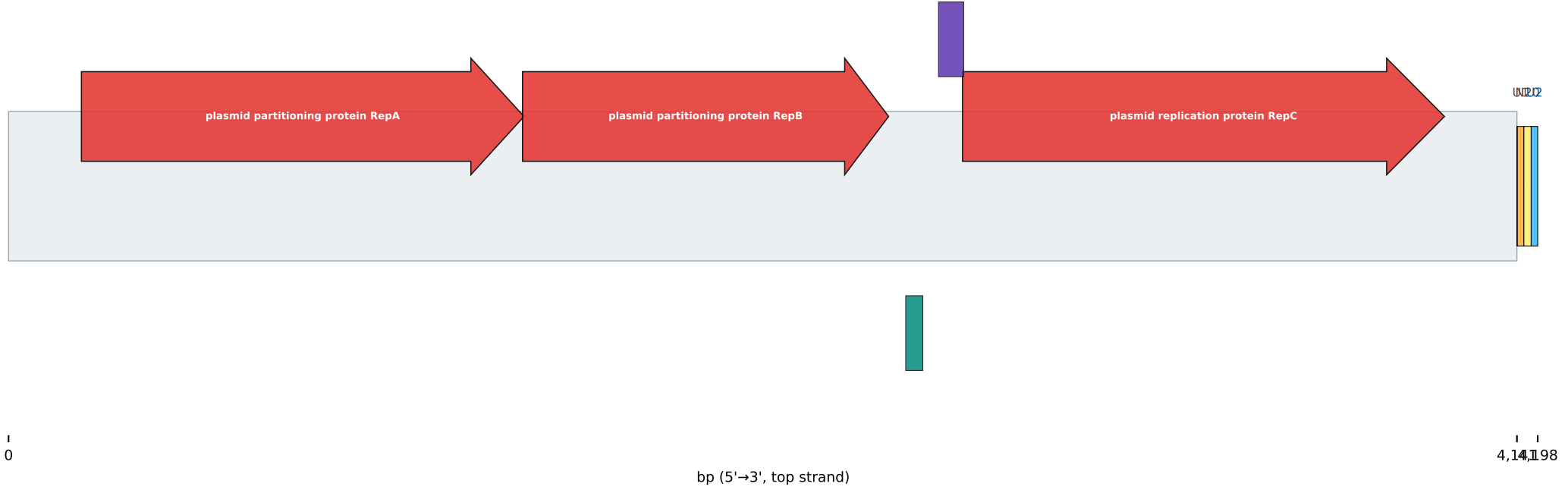

```
Tier / category:     PRIMARY (rank 26)
Rep family:          pAg
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5433
Barcode (N20):       ACGGTCCTAAGACATGAAGG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,022 bp (origin 2,965 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

B21_90_contig_3_pTi_Type_VII_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



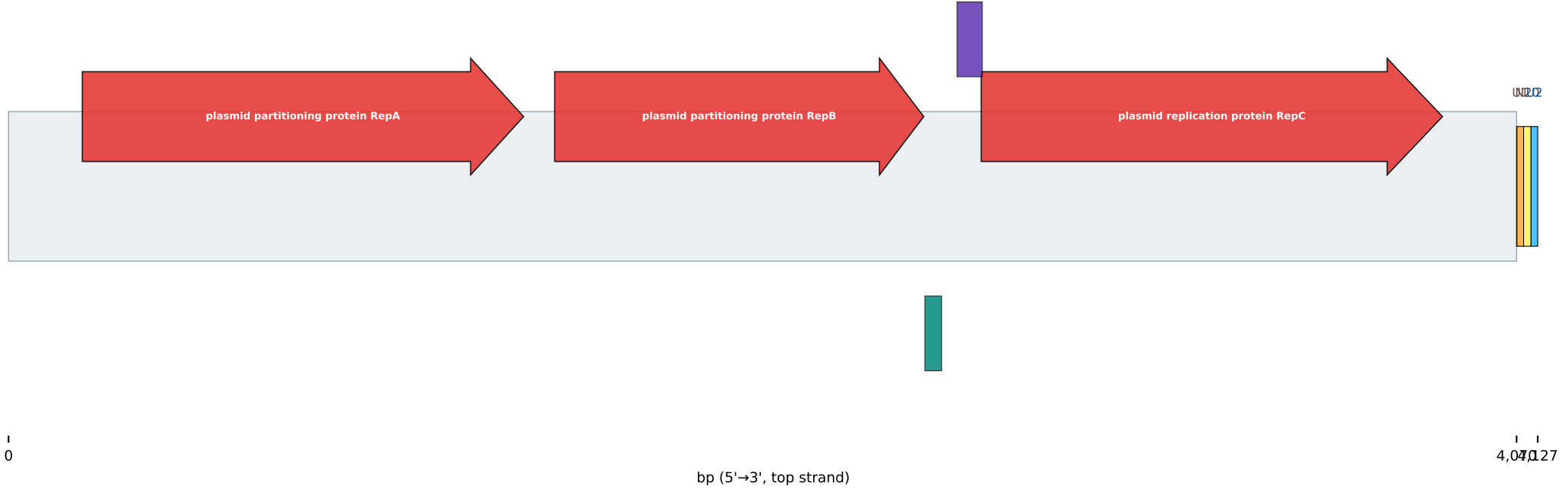

```
Tier / category:     PRIMARY (rank 12)
Rep family:          pTi Type VII
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.56
Barcode (N20):       GTTCGCTCTCCTGTTTGGCC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,198 bp (origin 4,141 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

B230_85_contig_5_pTi_Type_VIII.a_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



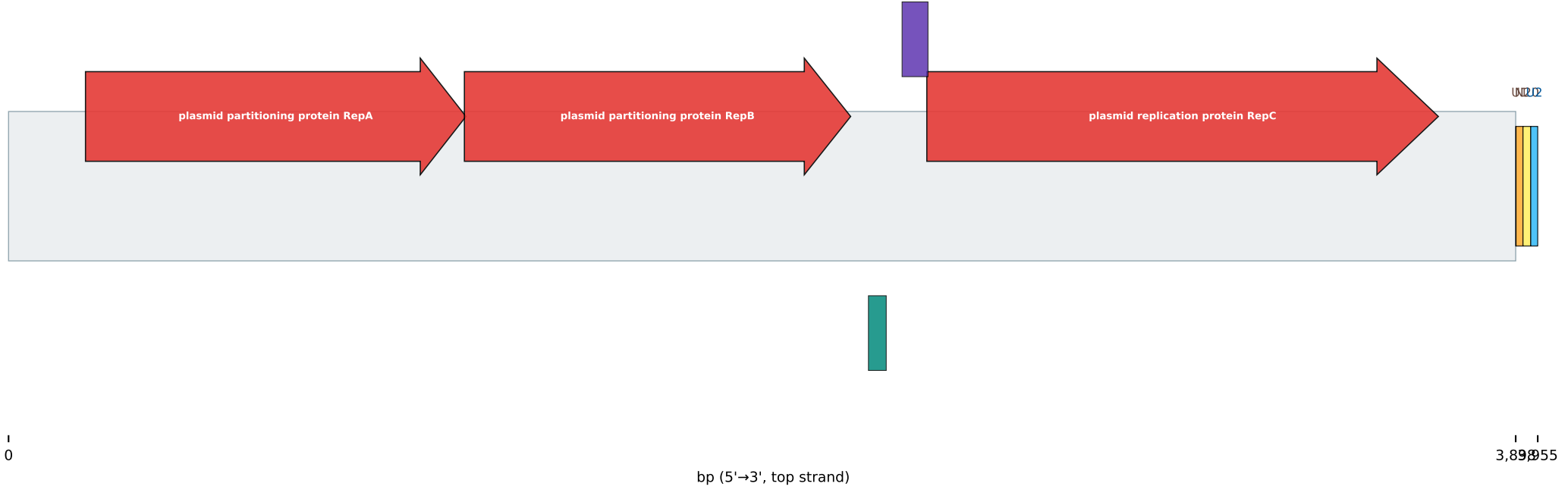

```
Tier / category:     PRIMARY (rank 10)
Rep family:          pTi Type VIII.a
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5442
Barcode (N20):       CACAGGCGAGCAATGAAACG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,127 bp (origin 4,070 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

B230_85_contig_6_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



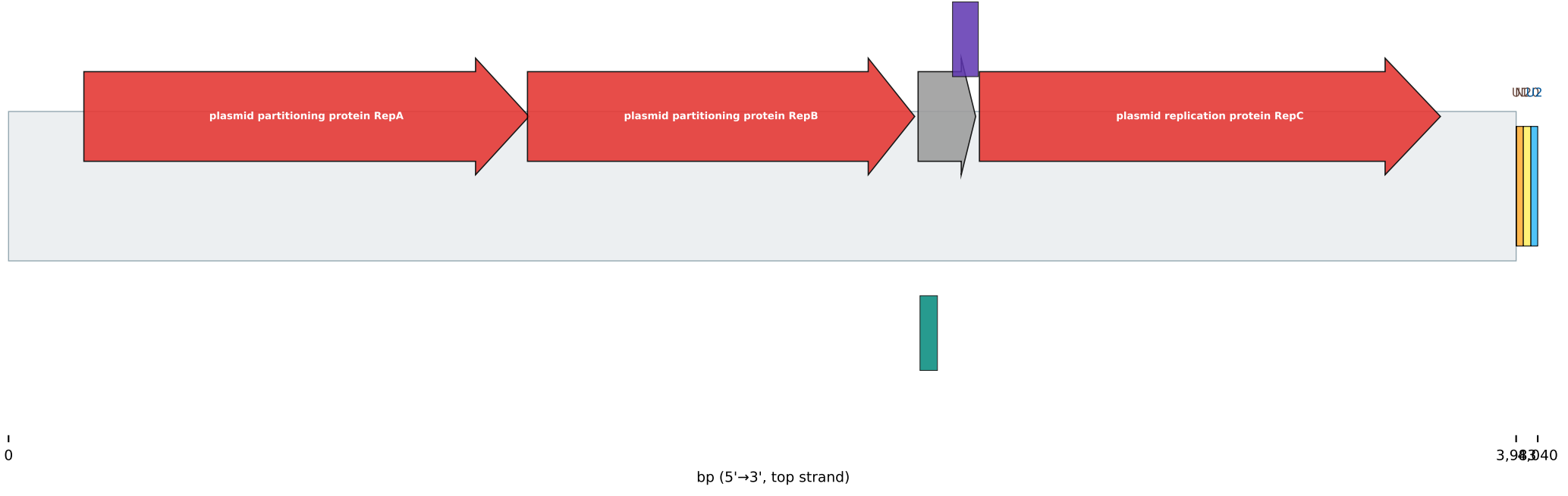

```
Tier / category:     PRIMARY (rank 32)
Rep family:          other_UniRef50_W6RJ16
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.559
Barcode (N20):       ATACGTGAGTTGATTGGACT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,955 bp (origin 3,898 + cassette 57)
Flags:               none
```

feature categories

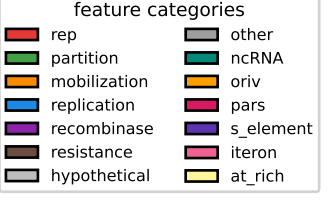
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

CA75_95_contig_2_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 4 CDS + 2 misc (.gbk)

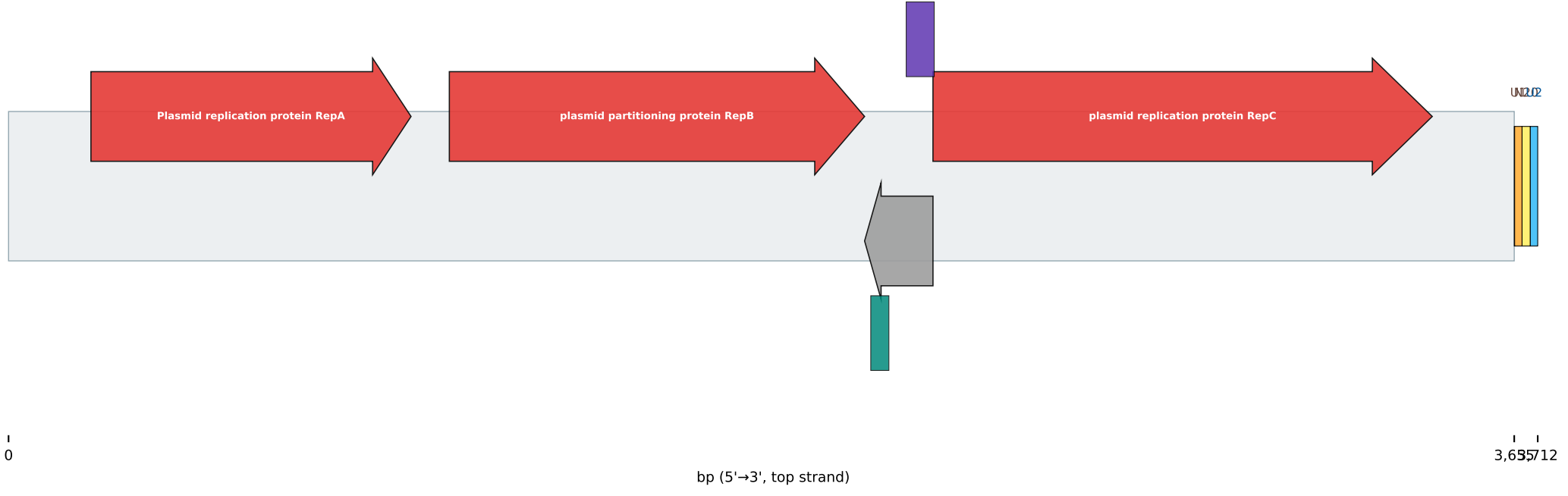



```
Tier / category:     PRIMARY (rank 30)
Rep family:          other_UniRef50_A0AA87PY09
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5722
Barcode (N20):       AACGTGTTGCCAGTCAGAAG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,040 bp (origin 3,983 + cassette 57)
Flags:               none
```



K599_CAICSX020000003.1_pRi_Type_I_repABC | Agrobacterium rhizogenes | repABC_primary | twist_synthesis

annotations: 4 CDS + 2 misc (.gbk)



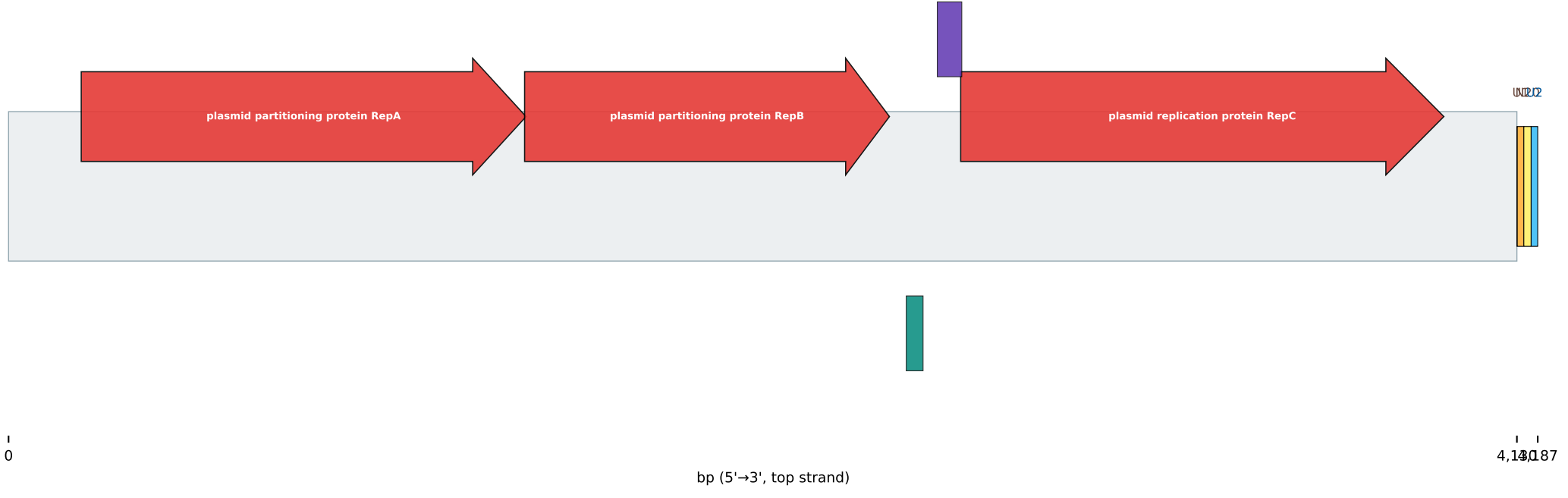

```
Tier / category:     PRIMARY (rank 20)
Rep family:          pRi Type I
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5357
Barcode (N20):       TCATAGGCGACGTGGATATC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,712 bp (origin 3,655 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

L51_94_contig_5_pTi_Type_VIII.b_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



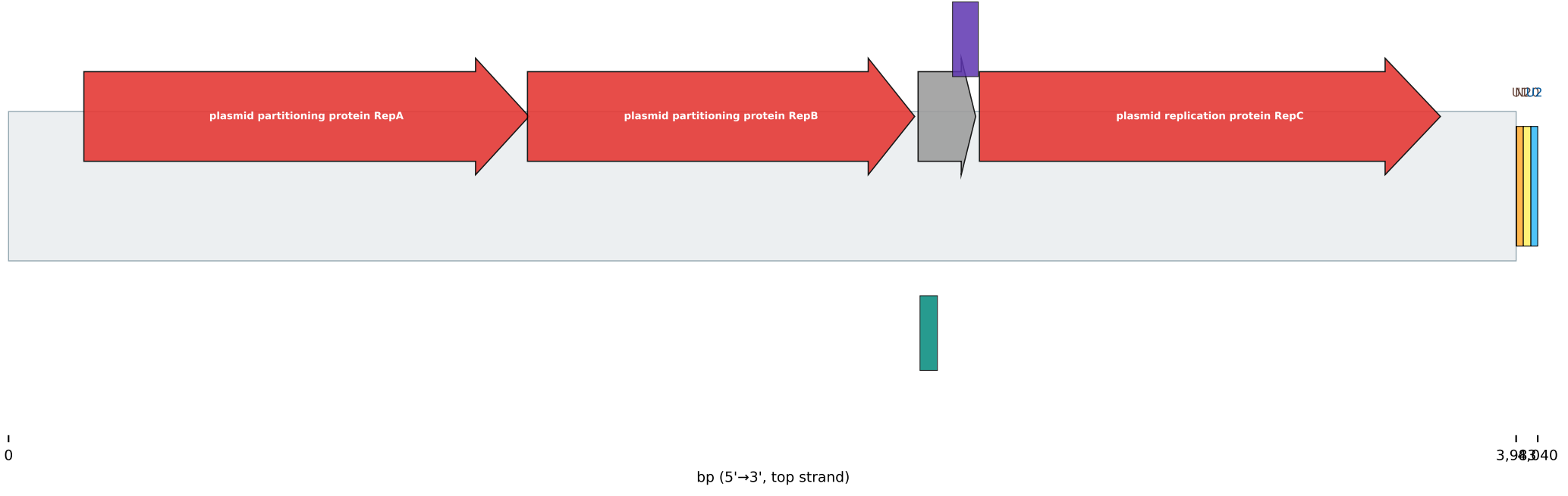

```
Tier / category:     PRIMARY (rank 11)
Rep family:          pTi Type VIII.b
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5518
Barcode (N20):       GCCATTTCGCTTCTGCATCC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,187 bp (origin 4,130 + cassette 57)
Flags:               none
```

feature categories

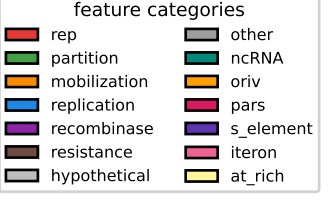
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

NCIB_8196_NZ_JAAMDI010000039.1_repABC | Agrobacterium rhizogenes | repABC_primary | twist_synthesis

annotations: 4 CDS + 2 misc (.gbk)

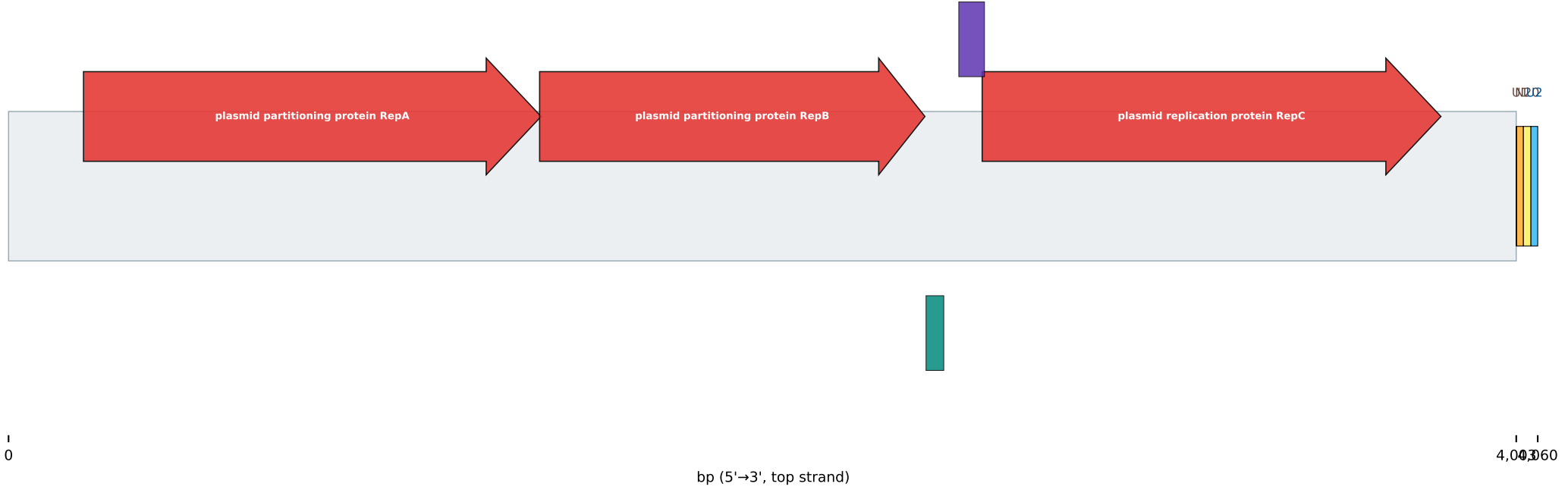



```
Tier / category:     PRIMARY (rank 29)
Rep family:          other_UniRef50_A0A1M3Q1S7
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5662
Barcode (N20):       TCCACGCACAGCACTCCTTT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,040 bp (origin 3,983 + cassette 57)
Flags:               none
```

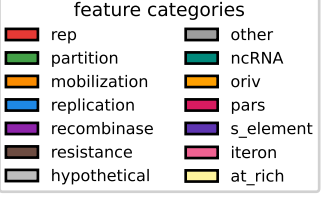


NCIB_8196_NZ_JAAMDI010000041.1_pRi_Type_II_repABC | Agrobacterium rhizogenes | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)

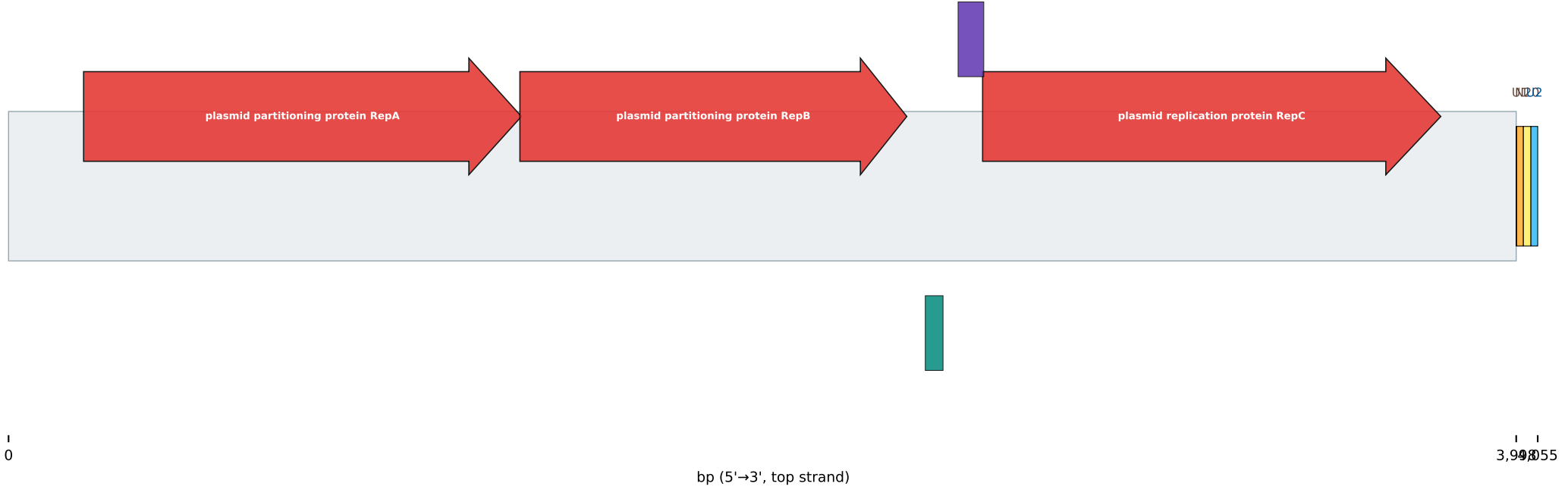



```
Tier / category:     PRIMARY (rank 22)
Rep family:          pRi Type II
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5526
Barcode (N20):       TGTTCGGTAATGGAATGACA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,060 bp (origin 4,003 + cassette 57)
Flags:               none
```



NCPPB_1641_NZ_LT009777.1_pTi_Type_XI_repABC | Agrobacterium vitis | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



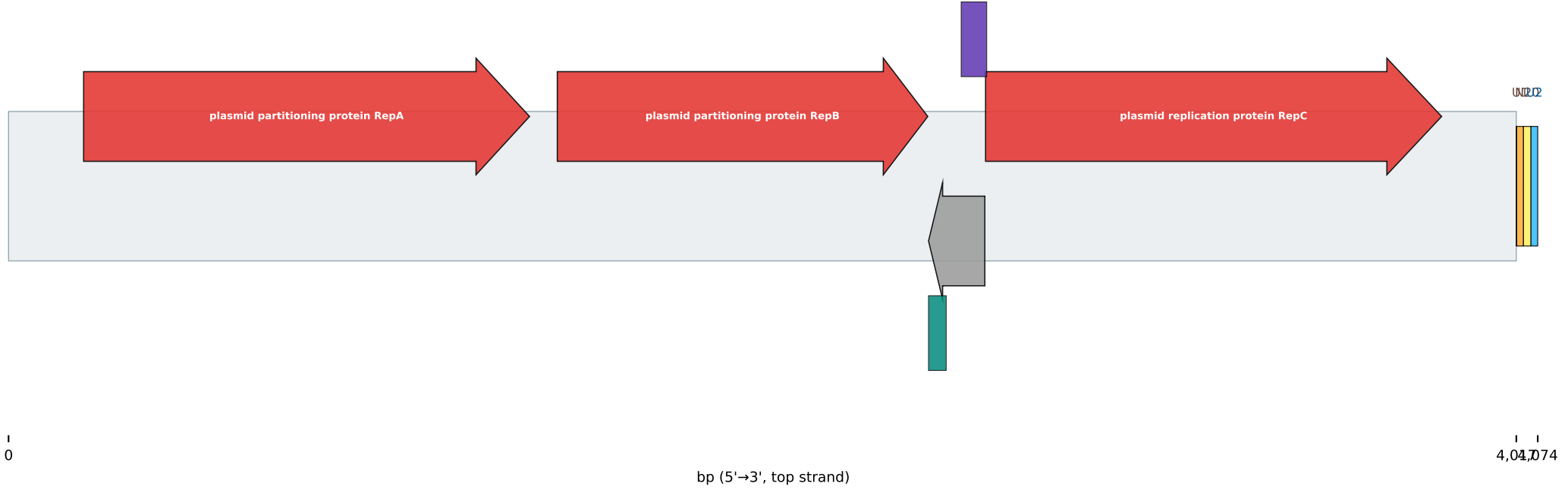

```
Tier / category:     PRIMARY (rank 7)
Rep family:          pTi Type XI
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.558
Barcode (N20):       GTCTCATTAAGTATCCACCT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,055 bp (origin 3,998 + cassette 57)
Flags:               none
```

feature categories

rep	other
partition	ncRNA
mobilization	oriv
replication	pars
recombinase	s_element
resistance	iteron
hypothetical	at_rich

Q15_94_contig_6_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 4 CDS + 2 misc (.gbk)



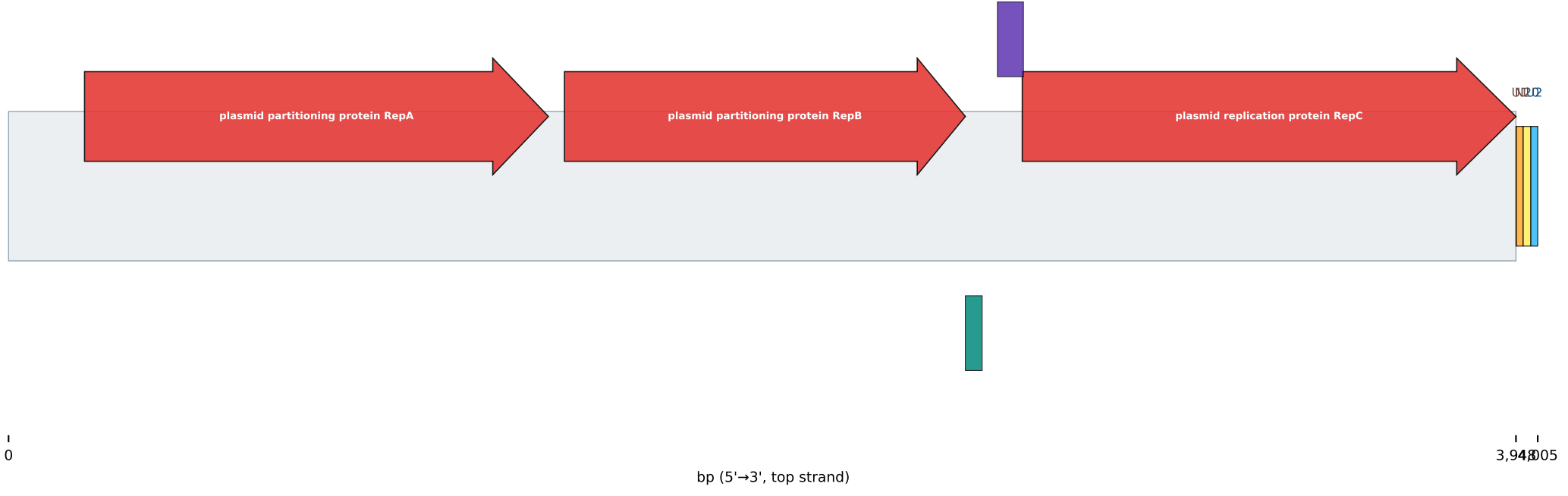

```
Tier / category:     PRIMARY (rank 31)
Rep family:          other_UniRef50_A0A7Z7LNC0
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5477
Barcode (N20):       TGGTACGTTCCGCTTAATGT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,074 bp (origin 4,017 + cassette 57)
Flags:               none
```

feature categories

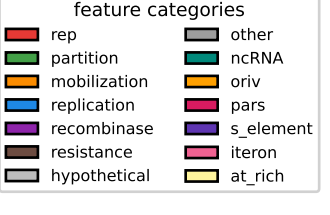
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Rhizobium_etli_CFN_42_CP000135.1_repABC | Rhizobium etli CFN 42 | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)

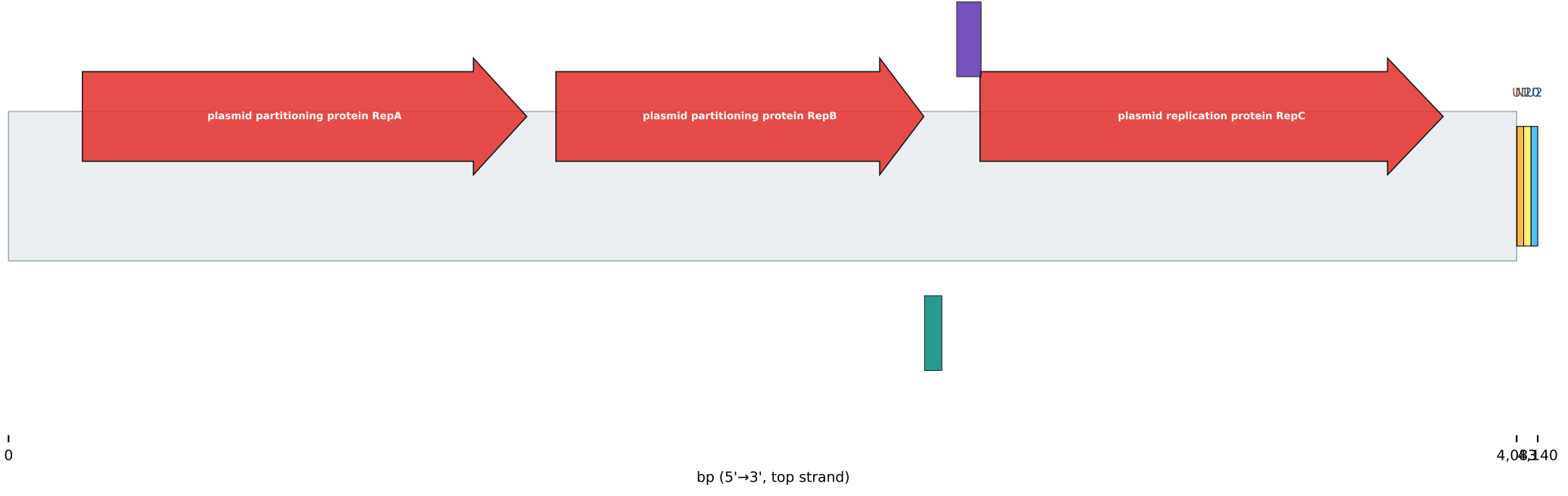



```
Tier / category:     PRIMARY (rank 33)
Rep family:          other_UniRef50_K0Q2E7
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5975
Barcode (N20):       CGTGCCTTCTCTTTGGAATC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,005 bp (origin 3,948 + cassette 57)
Flags:               none
```



S7_73_contig_3_pTi_Type_IX_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



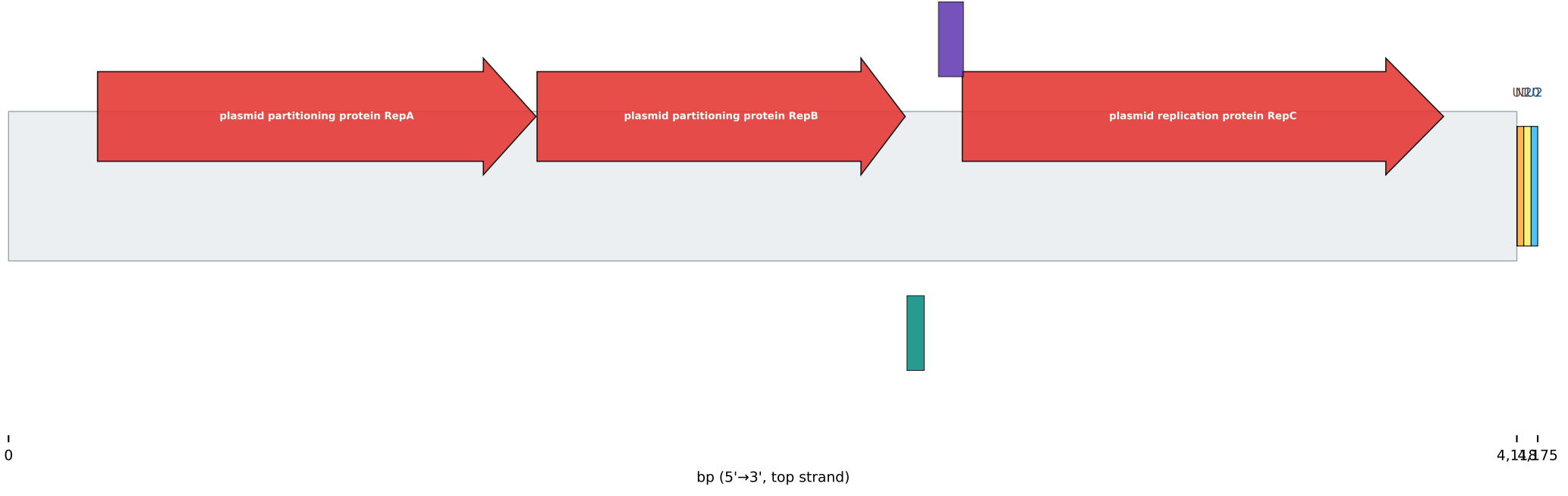

```
Tier / category:     PRIMARY (rank 14)
Rep family:          pTi Type IX
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5415
Barcode (N20):       TTCCGAGGTTTCCATCATGC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,140 bp (origin 4,083 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Sinorhizobium_fredii_NGR234_CP000874.1_repABC | Sinorhizobium fredii NGR234 | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



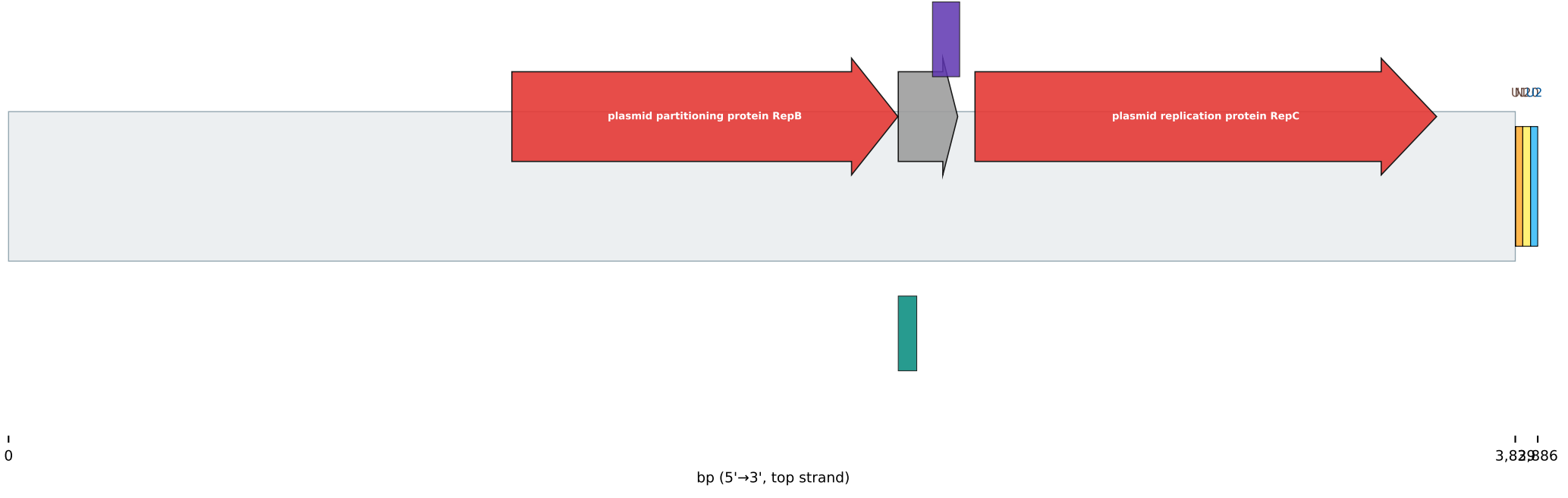

```
Tier / category:     PRIMARY (rank 3)
Rep family:          other_UniRef50_K0Q2E7
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.6146
Barcode (N20):       GCGTTATCGCGTTACGTCCG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,175 bp (origin 4,118 + cassette 57)
Flags:               none
```

feature categories

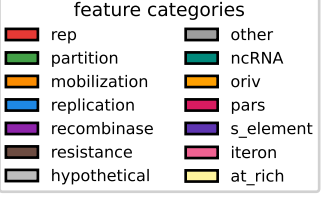
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Sinorhizobium_fredii_NGR234_U00090.2_repABC | Sinorhizobium fredii NGR234 | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)

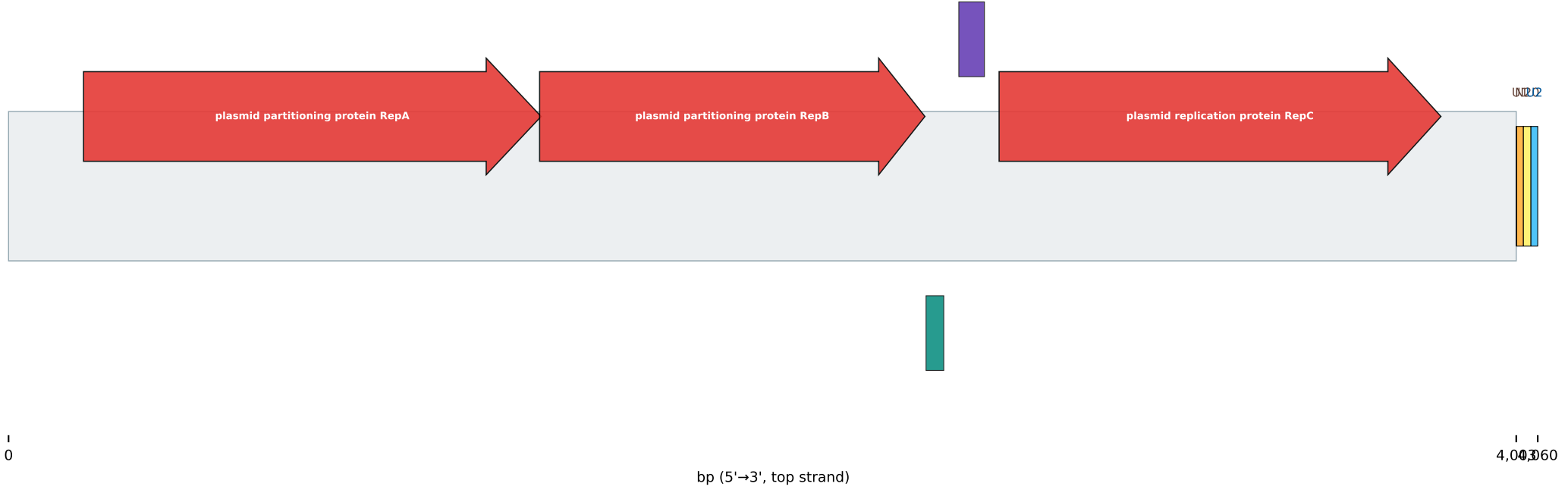



```
Tier / category:     PRIMARY (rank 1)
Rep family:          other_UniRef50_P05684
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5735
Barcode (N20):       CGTCACAGGCATGTCGAAAT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,886 bp (origin 3,829 + cassette 57)
Flags:               none
```



T155_95_NZ_JAAMCX010000012.1_pRi_Type_III_repABC | Agrobacterium rhizogenes | repABC_primary | twist_synthesis

annotations: 3 CDS + 2 misc (.gbk)



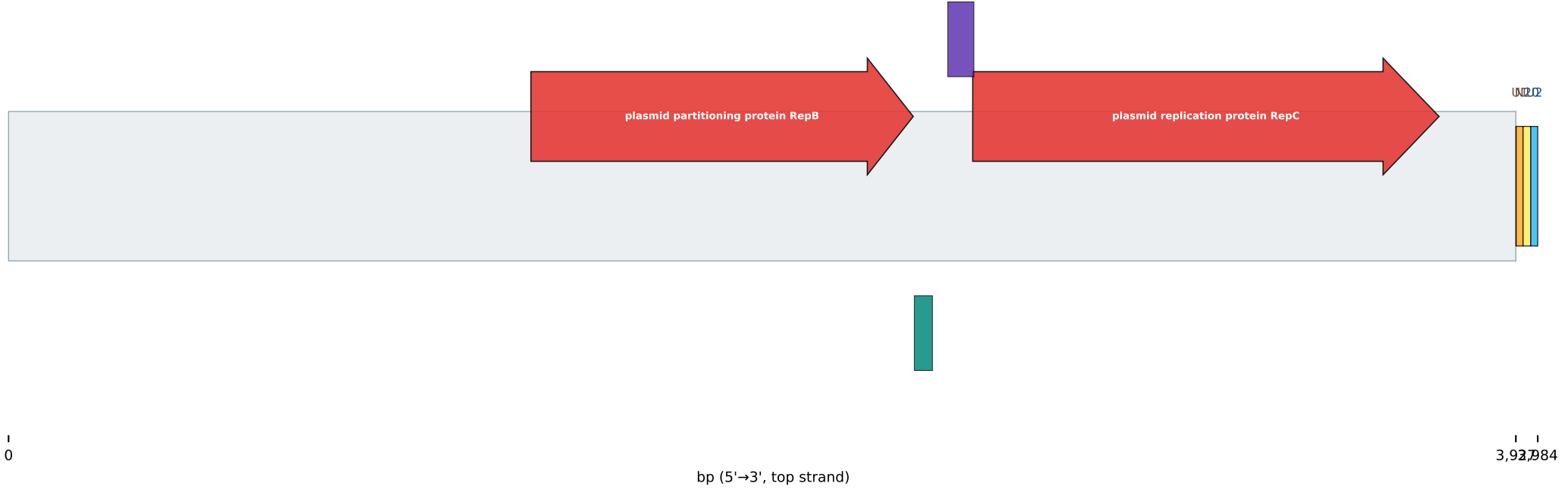

```
Tier / category:     PRIMARY (rank 21)
Rep family:          pRi Type III
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5501
Barcode (N20):       GCTGTGGAATGAATACTCTT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,060 bp (origin 4,003 + cassette 57)
Flags:               none
```

feature categories

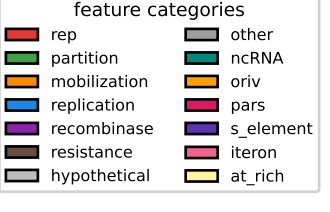
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

WS163_contig_4_repABC | Agrobacterium tumefaciens | repABC_primary | twist_synthesis

annotations: 2 CDS + 2 misc (.gbk)

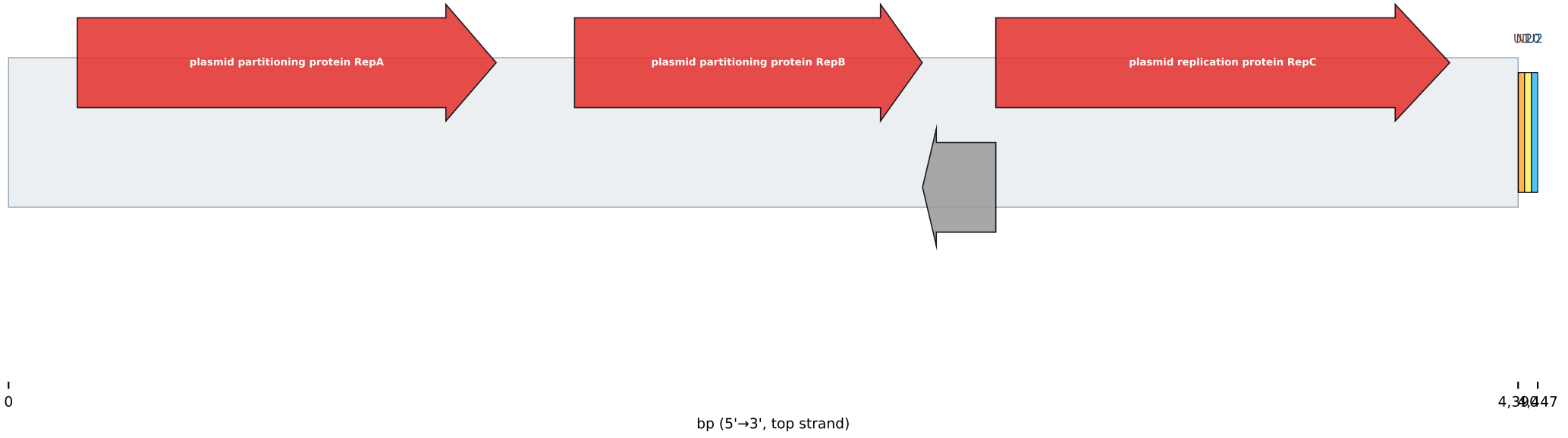



```
Tier / category:     PRIMARY (rank 2)
Rep family:          other_UniRef50_A0A7Z7LNC0
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5867
Barcode (N20):       GTTTCCCTGGGCGGTCTATG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,984 bp (origin 3,927 + cassette 57)
Flags:               none
```

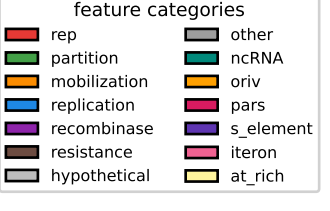


1D1524_plasmid2_pTi_Type_IV.a_repABC | Agrobacterium tumefaciens | repABC_alt_cloning | genomic_pcr_amplification

annotations: 4 CDS + 0 misc (.gbk)

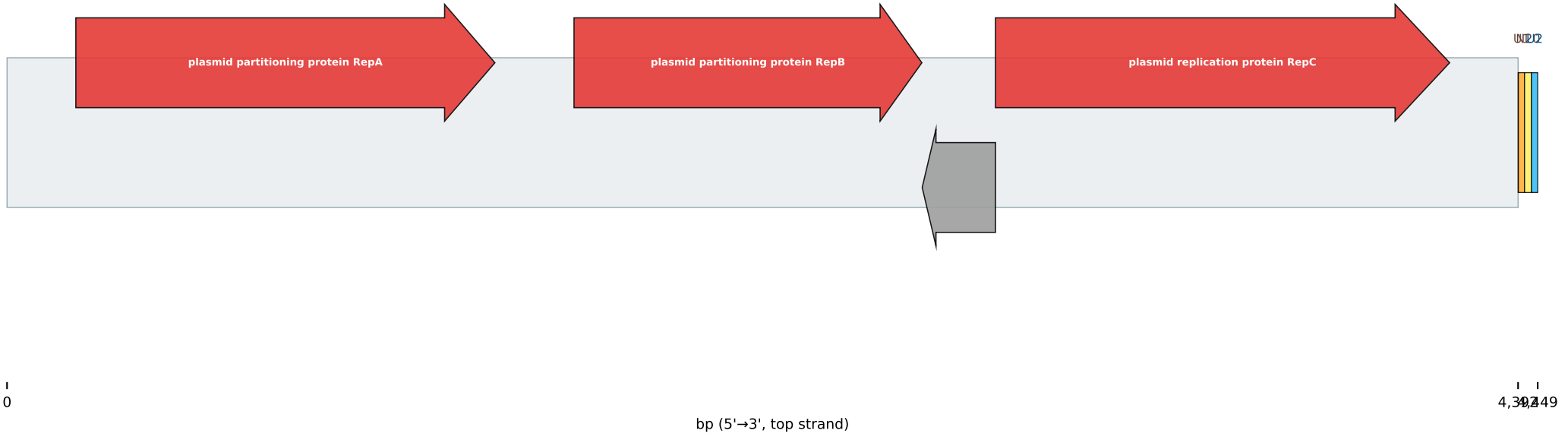



```
Tier / category:     PRIMARY (rank 8)
Rep family:          pTi Type IV.a
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.561
Barcode (N20):       GAGTTACAGCACTAGGAAGG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,447 bp (origin 4,390 + cassette 57)
Flags:               none
```



AMS008_plasmid3_pTi_Type_IV.b_repABC | Agrobacterium tumefaciens | repABC_alt_cloning | genomic_pcr_amplification

annotations: 4 CDS + 0 misc (.gbk)



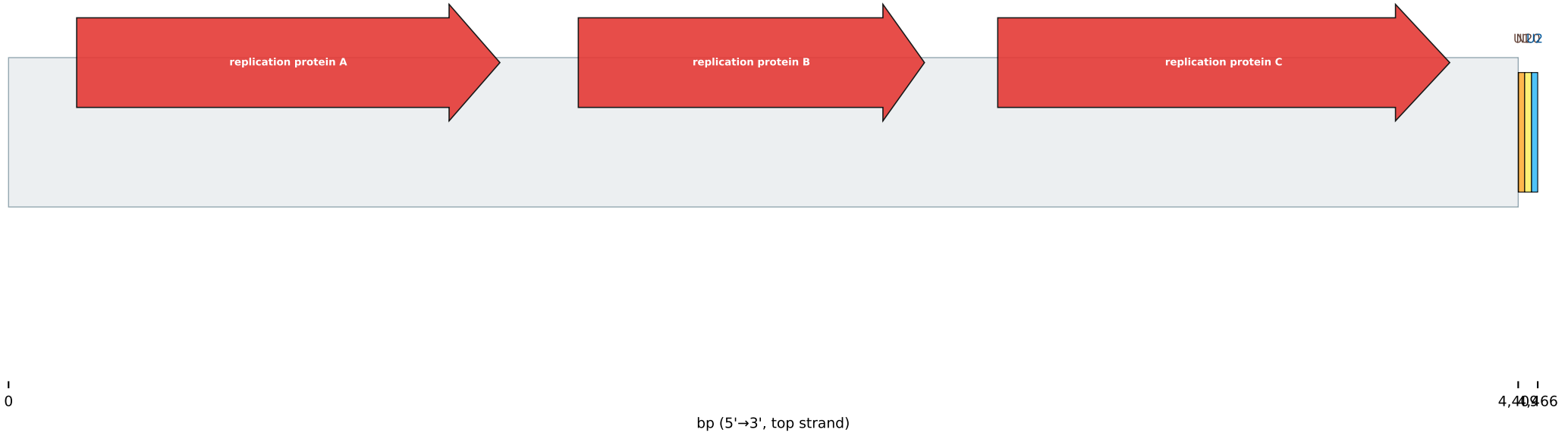

```
Tier / category:     PRIMARY (rank 9)
Rep family:          pTi Type IV.b
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5606
Barcode (N20):       AGAAAGAGAGTGAACTTCGC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,449 bp (origin 4,392 + cassette 57)
Flags:               none
```

feature categories

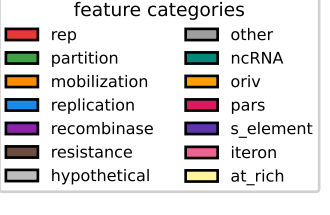
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Atu_K84_pAtK84b_CP000630.1_repABC | Agrobacterium radiobacter | repABC_alt_cloning | genomic_pcr_amplification

annotations: 3 CDS + 0 misc (.gbk)

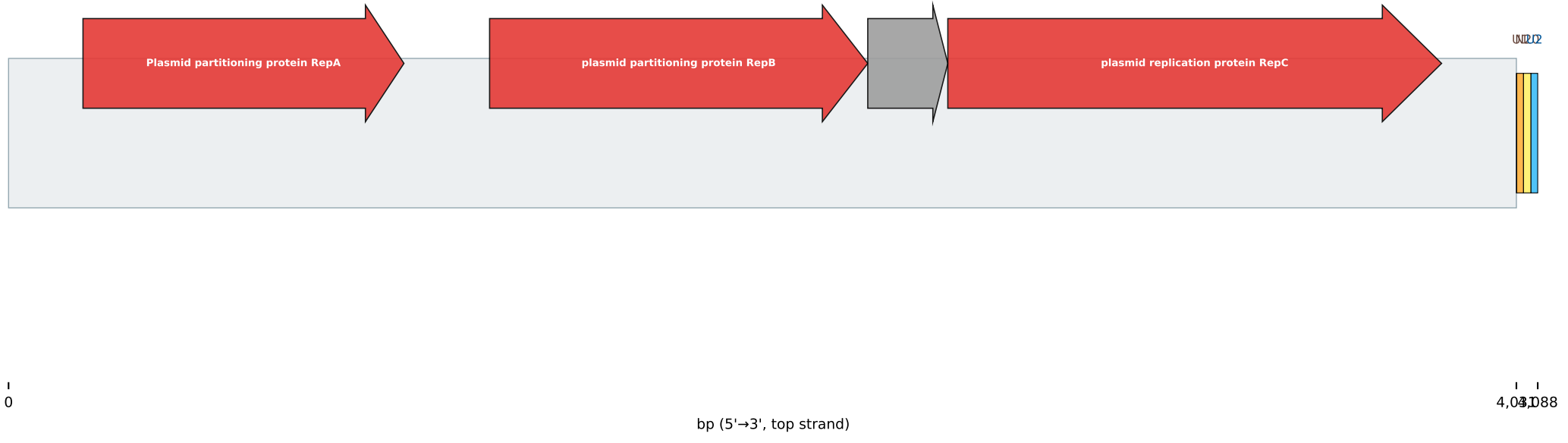



```
Tier / category:     PRIMARY (rank 25)
Rep family:          pAt
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5611
Barcode (N20):       ATTTGCGCGTGACATCTATC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,466 bp (origin 4,409 + cassette 57)
Flags:               none
```

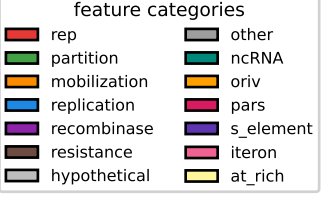


C58_AE007871.2_pTi_Type_I.a_repABC | Agrobacterium fabrum | repABC_alt_cloning | genomic_pcr_amplification

annotations: 4 CDS + 0 misc (.gbk)

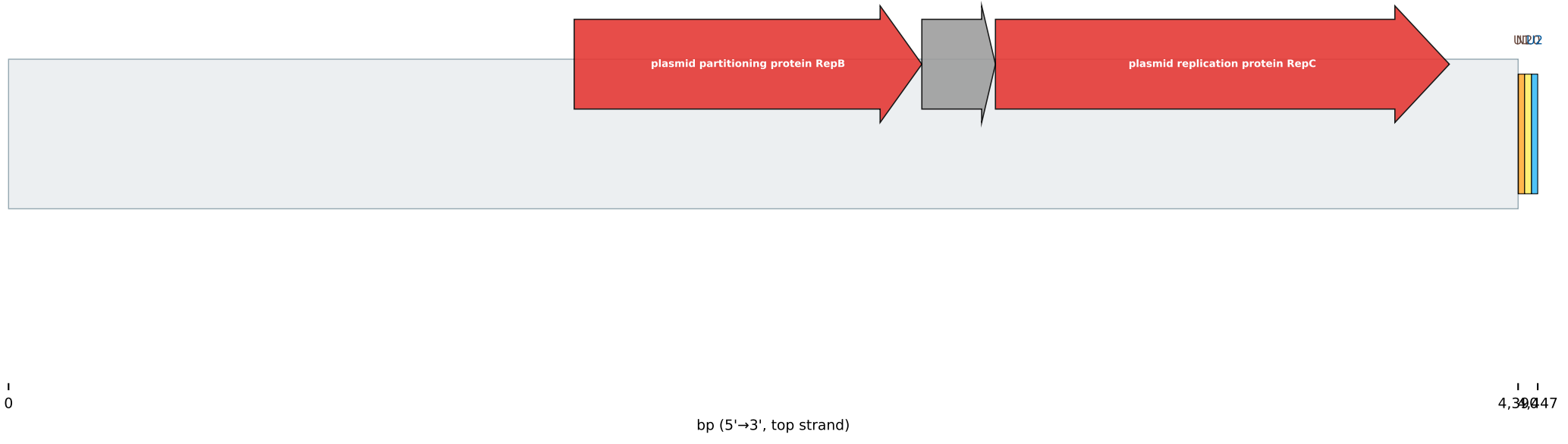



```
Tier / category:     PRIMARY (rank 6)
Rep family:          pTi Type I.a
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5644
Barcode (N20):       TAGAATTTGCGGTTCCACGC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,088 bp (origin 4,031 + cassette 57)
Flags:               none
```



CA75_95_contig_3_pTi_Type_IV.c_repABC | Agrobacterium tumefaciens | repABC_alt_cloning | genomic_pcr_amplification

annotations: 3 CDS + 0 misc (.gbk)



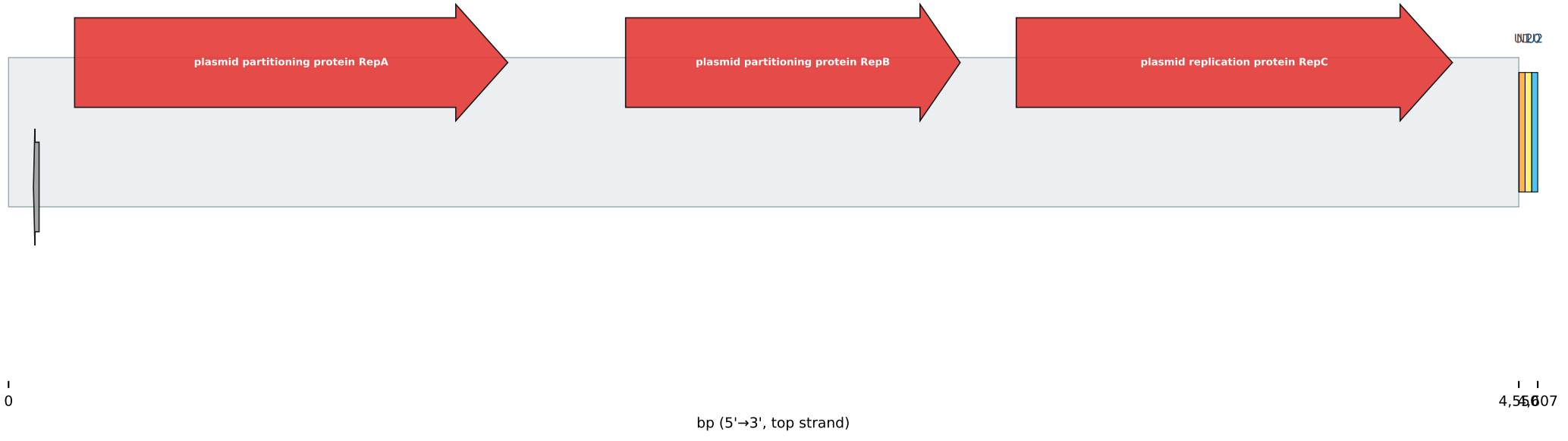

```
Tier / category:     PRIMARY (rank 19)
Rep family:          pTi Type IV.c
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5656
Barcode (N20):       TCCAATATCAGCCGGCACTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,447 bp (origin 4,390 + cassette 57)
Flags:               none
```

feature categories

- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

CG474_CG474_3_pTi_Type_I.c_repABC | Agrobacterium tumefaciens | repABC_alt_cloning | genomic_pcr_amplification

annotations: 4 CDS + 0 misc (.gbk)



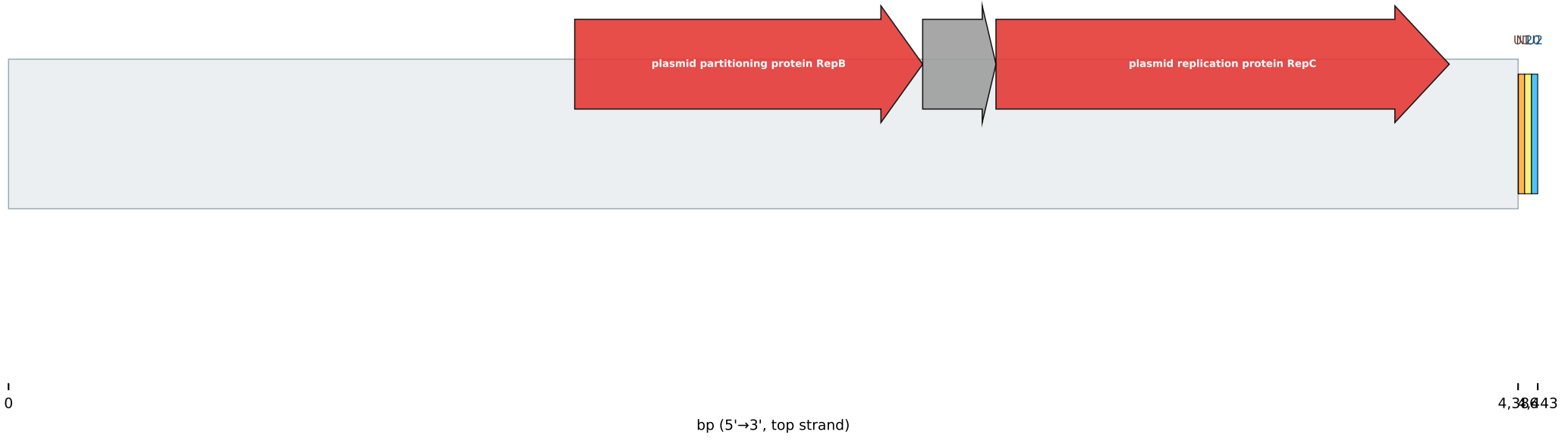

```
Tier / category:     PRIMARY (rank 16)
Rep family:          pTi Type I.c
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5618
Barcode (N20):       ACTCGATCTCCTAGCGAGTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,607 bp (origin 4,550 + cassette 57)
Flags:               none
```

feature categories

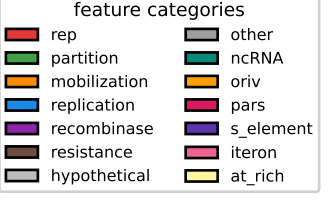
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

Q15_94_contig_5_pTi_Type_VI_repABC | Agrobacterium tumefaciens | repABC_alt_cloning | genomic_pcr_amplification

annotations: 3 CDS + 0 misc (.gbk)



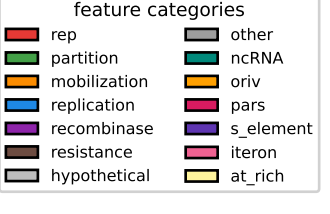

```
Tier / category:     PRIMARY (rank 18)
Rep family:          pTi Type VI
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5618
Barcode (N20):       GATGATATGCCCTCTCCTTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,443 bp (origin 4,386 + cassette 57)
Flags:               none
```



1054768854 | Sphingobium sp. EP60837 | diversity_PLSDB | twist_synthesis

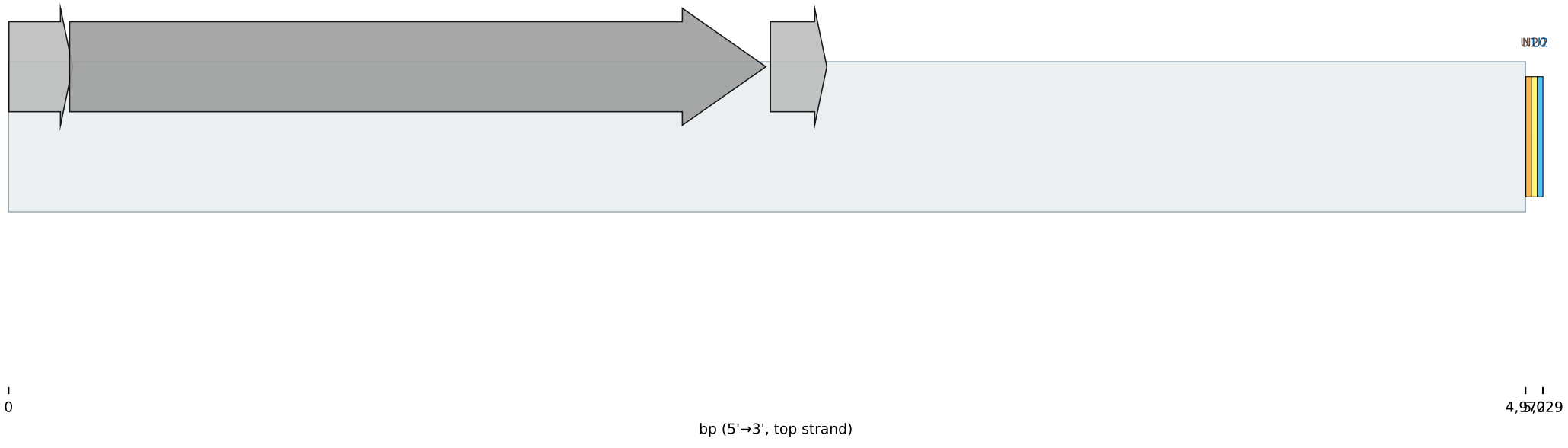
annotations: 2 CDS + 0 misc (.gb (NZ_CP015988.1))

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RPA
Taxonomy:            Alphaproteobacteria / Sphingomonadaceae
GC content:          0.5870140965399402
Barcode (N20):       AGCAGAGCGGAACAATCCTA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    5,214 bp (origin 5,157 + cassette 57)
Flags:               OVERSIZE >5 kb (Ian to re-review)
```

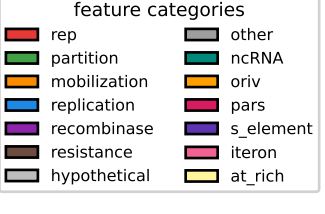


1327621529 | Salmonella enterica | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_CP019203.1))

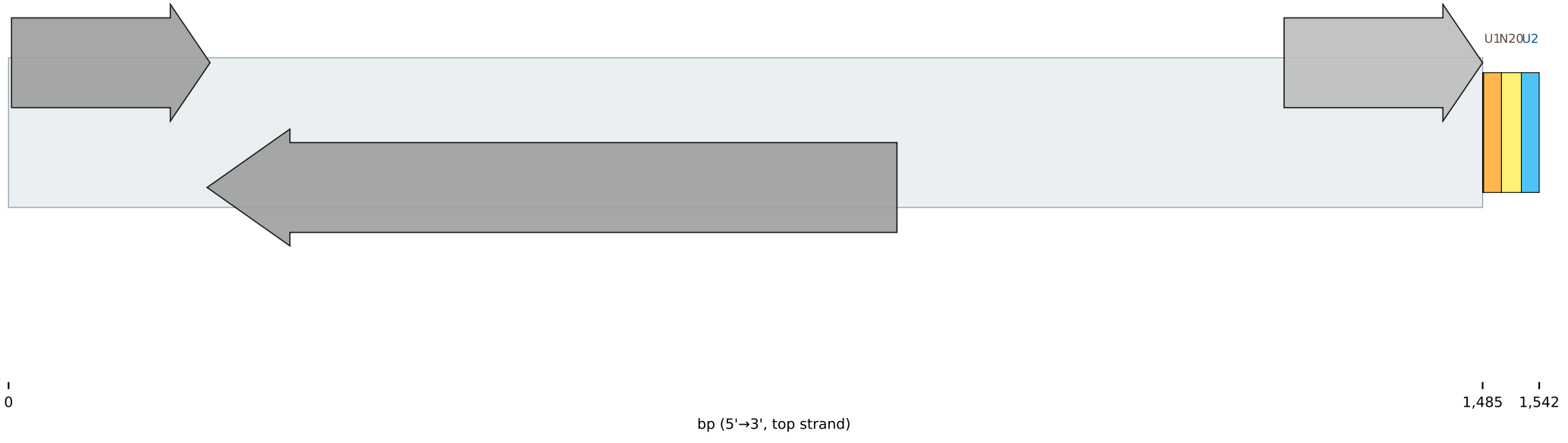



```
Tier / category:     Tier 3: Cross-phylum BHR
Rep family:          REP_ORF2-G2P__RCR
Taxonomy:            Gammaproteobacteria / Enterobacteriaceae
GC content:          0.5231263815796856
Barcode (N20):       TGAGTTCAAGGGCGGTAGTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    5,029 bp (origin 4,972 + cassette 57)
Flags:               OVERSIZE >5 kb (Ian to re-review)
```

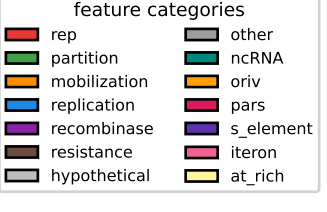


1441331292 | Sulfitobacter sp. SK025 | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_CP025809.1))

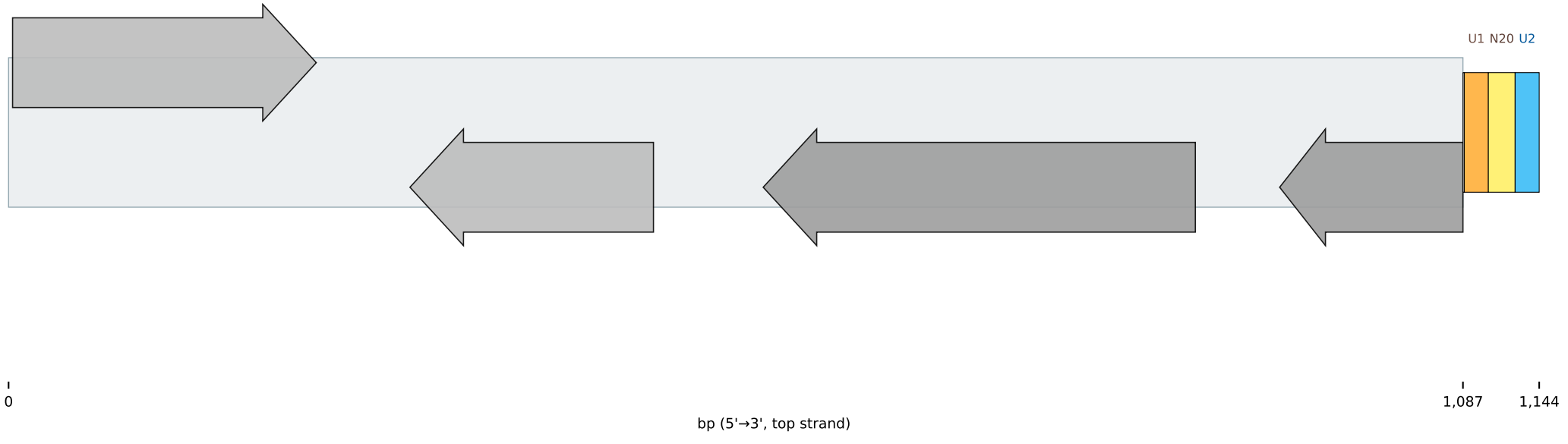



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          DnaA_N
Taxonomy:            Alphaproteobacteria / Roseobacteraceae
GC content:          0.5970530917234622
Barcode (N20):       GGCAATACGTTCTTGACACC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,542 bp (origin 1,485 + cassette 57)
Flags:               none
```



1586008349 | Lichenihabitans psoromatis | diversity_PLSDB | twist_synthesis

annotations: 4 CDS + 0 misc (.gb (NZ_CP036517.1))




```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          PRK13824.1
Taxonomy:            Alphaproteobacteria / Lichenihabitantaceae
GC content:          0.6050201586682273
Barcode (N20):       TCGTCATCAGGTCCTTCGTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,144 bp (origin 1,087 + cassette 57)
Flags:               none
```

feature categories

rep	other
partition	ncRNA
mobilization	oriv
replication	pars
recombinase	s_element
resistance	iteron
hypothetical	at_rich

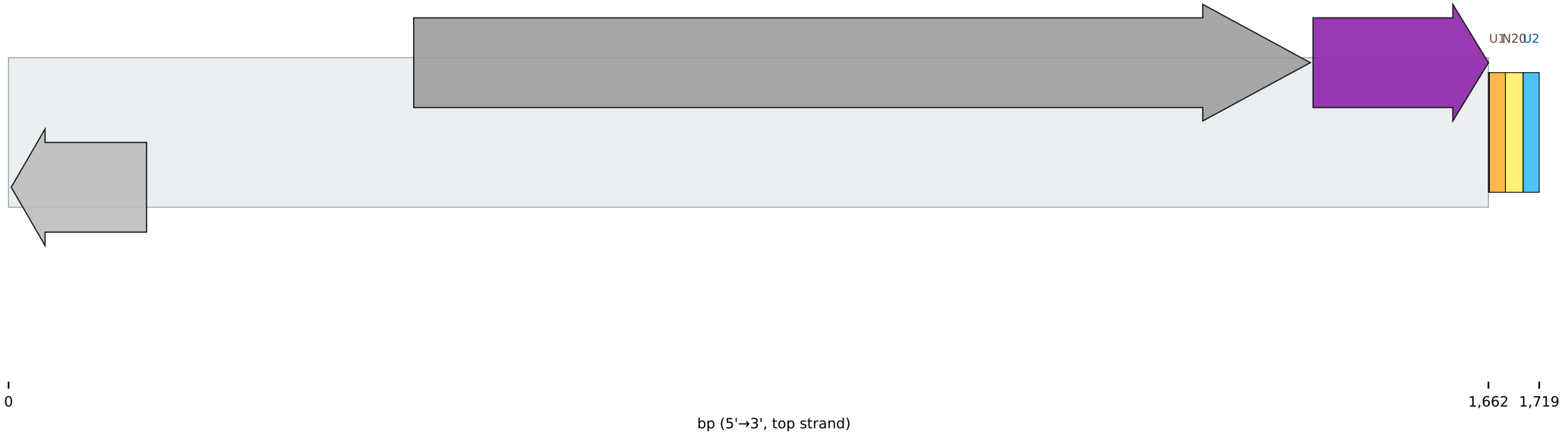
1752200521 | Pseudomonas oryzihabitans | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NZ_CP044072.1))

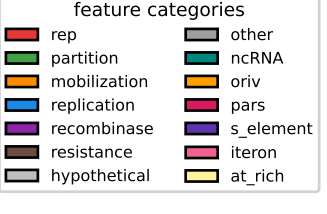
```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          repM_Acin
Taxonomy:            Gammaproteobacteria / Pseudomonadaceae
GC content:          0.5865734816418939
Barcode (N20):       CAGCGAGTCATTCGAATATT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,916 bp (origin 3,859 + cassette 57)
Flags:               none
```

1817008522 | Mycolicibacterium aubagnense | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_AP022578.1))

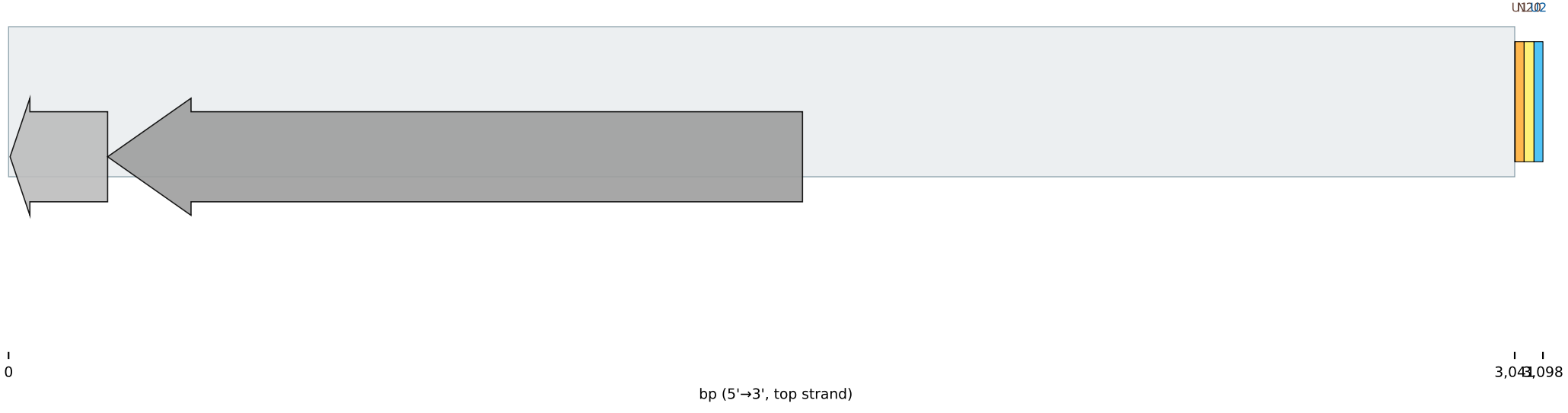



```
Tier / category:     Tier 4: Phylum-exclusive
Rep family:          rep_init_RepSA
Taxonomy:            Actinomycetes / Mycobacteriaceae
GC content:          0.6488044526678852
Barcode (N20):       GTGACTTTGTTAAGTCACGA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,719 bp (origin 1,662 + cassette 57)
Flags:               none
```



1819419980 | Caballeronia sp. SBC2 | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_CP049320.1))

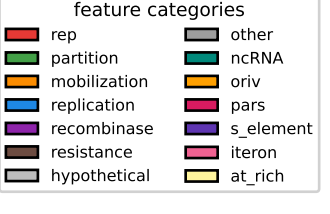



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep3_C
Taxonomy:            Betaproteobacteria / Burkholderiaceae
GC content:          0.5913467527132502
Barcode (N20):       ATACCCGATGGTAACTTACG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,098 bp (origin 3,041 + cassette 57)
Flags:               none
```

1848403693 | Aureimonas ureilytica | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_LC066381.1))

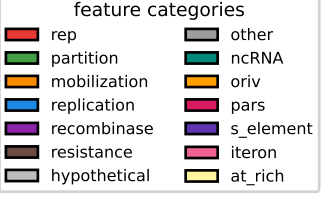
```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          repM_Acin
Taxonomy:            Alphaproteobacteria / Aurantimonadaceae
GC content:          0.593439662955161
Barcode (N20):       AGGAATGCTGACGACGATCA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    6,057 bp (origin 6,000 + cassette 57)
Flags:               OVERSIZE >5 kb (Ian to re-review)
```



1848403714 | Aureimonas sp. D3 | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NZ_LC066394.1))

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          repM_Acin
Taxonomy:            Alphaproteobacteria / Aurantimonadaceae
GC content:          0.5769774011299436
Barcode (N20):       CCAACATCTACCAATTGCGA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    6,057 bp (origin 6,000 + cassette 57)
Flags:               OVERSIZE >5 kb (Ian to re-review)
```



1848403722 | Aureimonas sp. N4 | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NZ_LC066398.1))

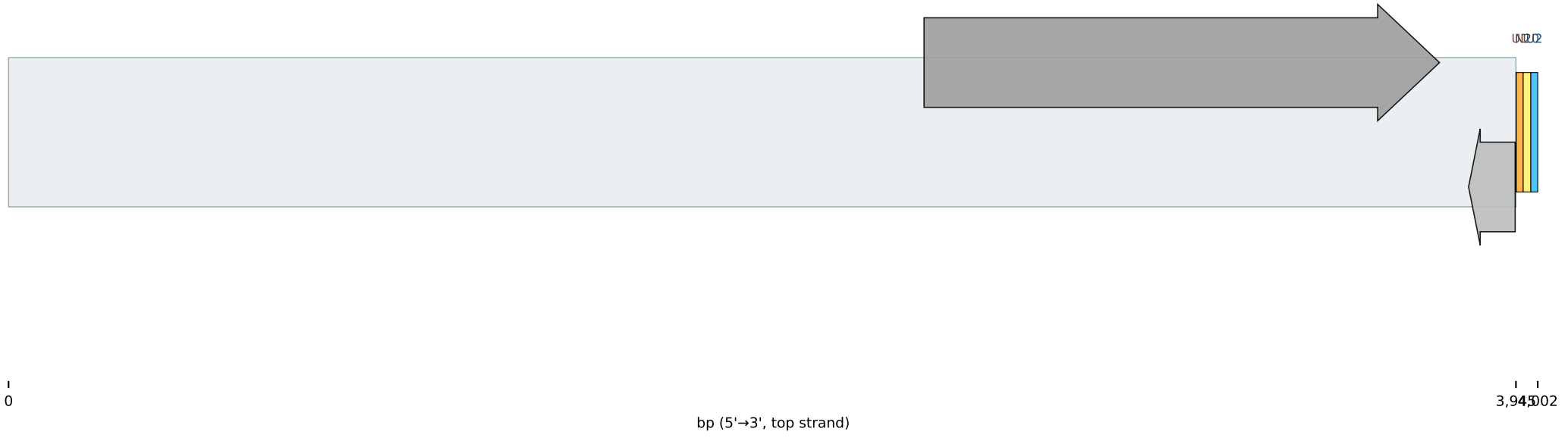
```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          repM_Acin
Taxonomy:            Alphaproteobacteria / Aurantimonadaceae
GC content:          0.5767521712785296
Barcode (N20):       CCTGTGCCTAAGTGTCACTA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    6,057 bp (origin 6,000 + cassette 57)
Flags:               OVERSIZE >5 kb (Ian to re-review)
```

feature categories

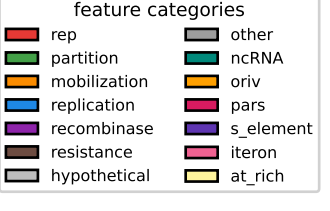
rep	other
partition	ncRNA
mobilization	oriv
replication	pars
recombinase	s_element
resistance	iteron
hypothetical	at_rich

1862425362 | Janthinobacterium lividum | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_CP049829.1))

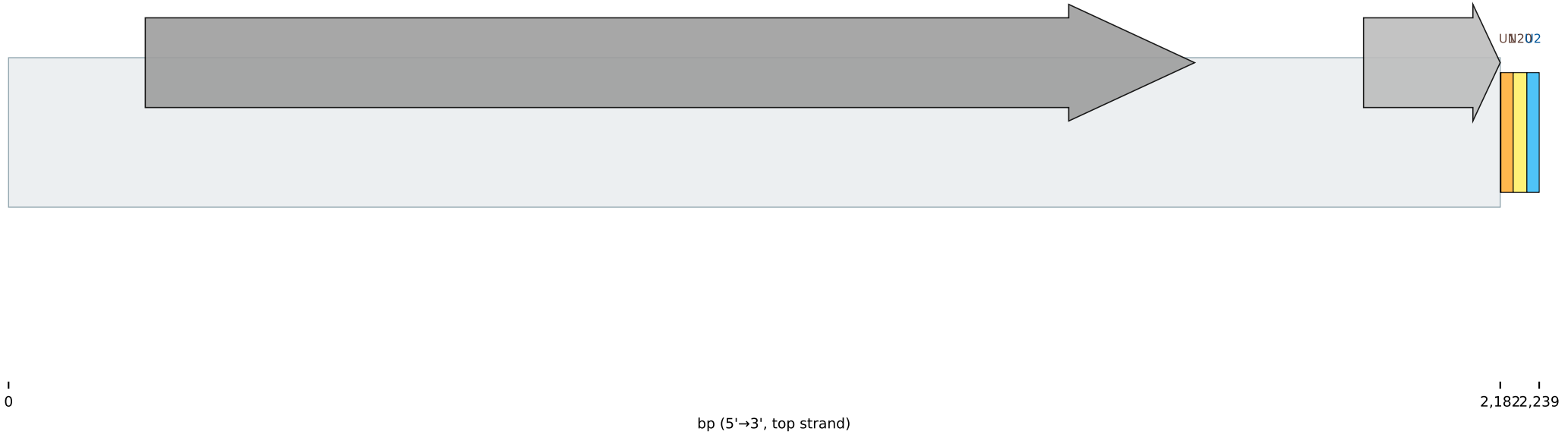



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep3_C
Taxonomy:            Betaproteobacteria / Oxalobacteraceae
GC content:          0.5721151555743627
Barcode (N20):       CGGAGTCTGTTTACGTATTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    4,002 bp (origin 3,945 + cassette 57)
Flags:               none
```

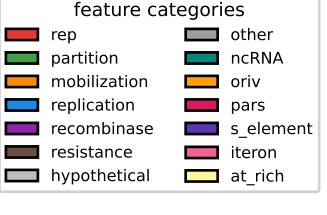


1875166974 | Limosilactobacillus mucosae | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_CP058955.1))

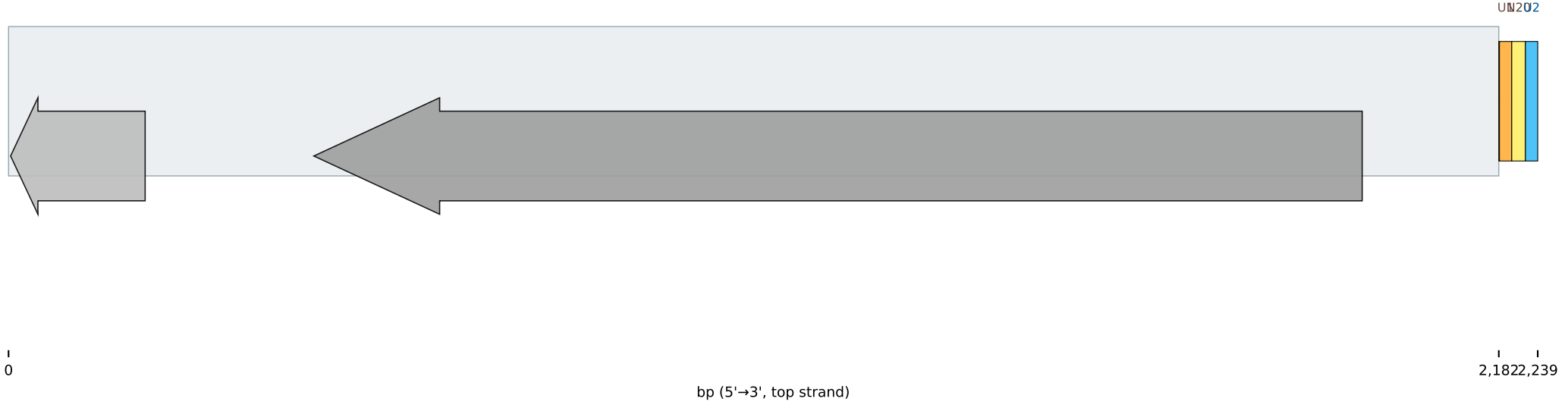



```
Tier / category:     Tier 4: Phylum-exclusive
Rep family:          RepA_N
Taxonomy:            Bacilli / Lactobacillaceae
GC content:          0.4427031871136745
Barcode (N20):       TTGGACAGATTCTCCGCGAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,239 bp (origin 2,182 + cassette 57)
Flags:               none
```

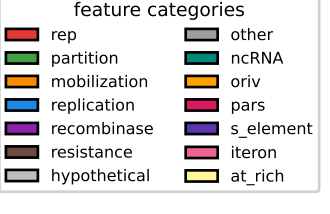


1875166978 | Limosilactobacillus mucosae | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_CP058956.1))

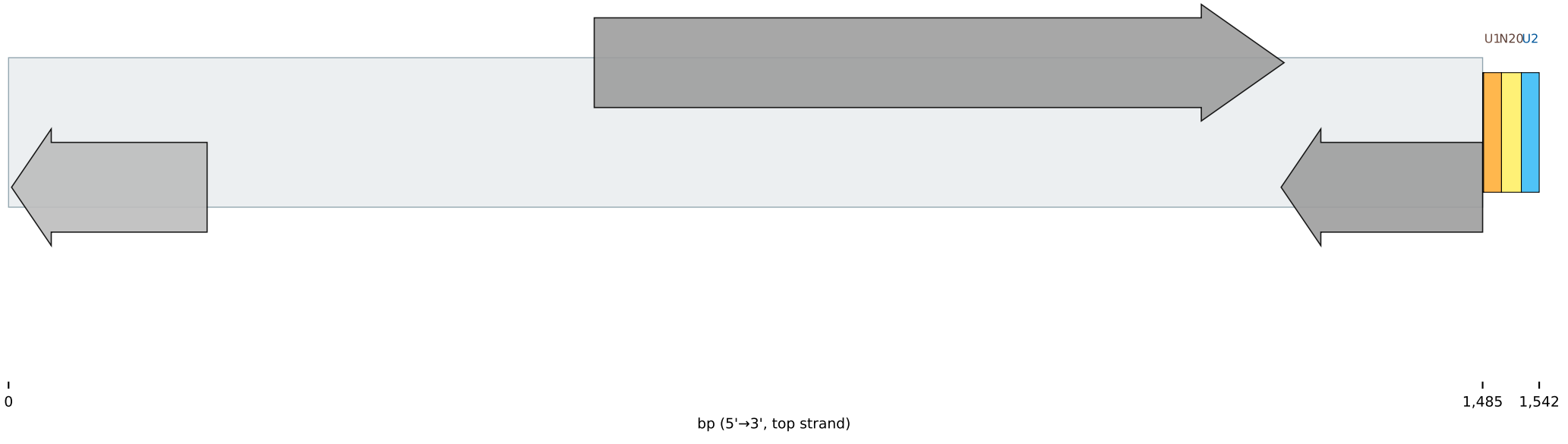



```
Tier / category:     Tier 4: Phylum-exclusive
Rep family:          RepA_N
Taxonomy:            Bacilli / Lactobacillaceae
GC content:          0.4427412497414565
Barcode (N20):       AAAGGCTAACAGTCAGGCAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,239 bp (origin 2,182 + cassette 57)
Flags:               none
```

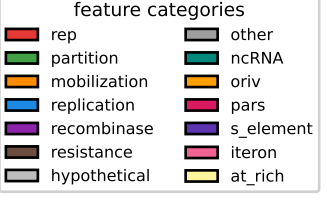


1878874587 | Sulfitobacter pontiacus | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_CP049346.1))



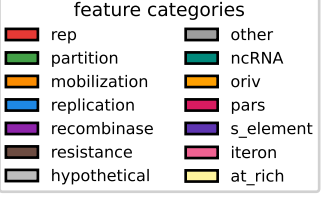

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          DnaA_N
Taxonomy:            Alphaproteobacteria / Roseobacteraceae
GC content:          0.5945672214515226
Barcode (N20):       CGCGAGTAATGACCCGACGT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,542 bp (origin 1,485 + cassette 57)
Flags:               none
```



1883401799 | Enterobacter hormaechei | diversity_PLSDB | twist_synthesis

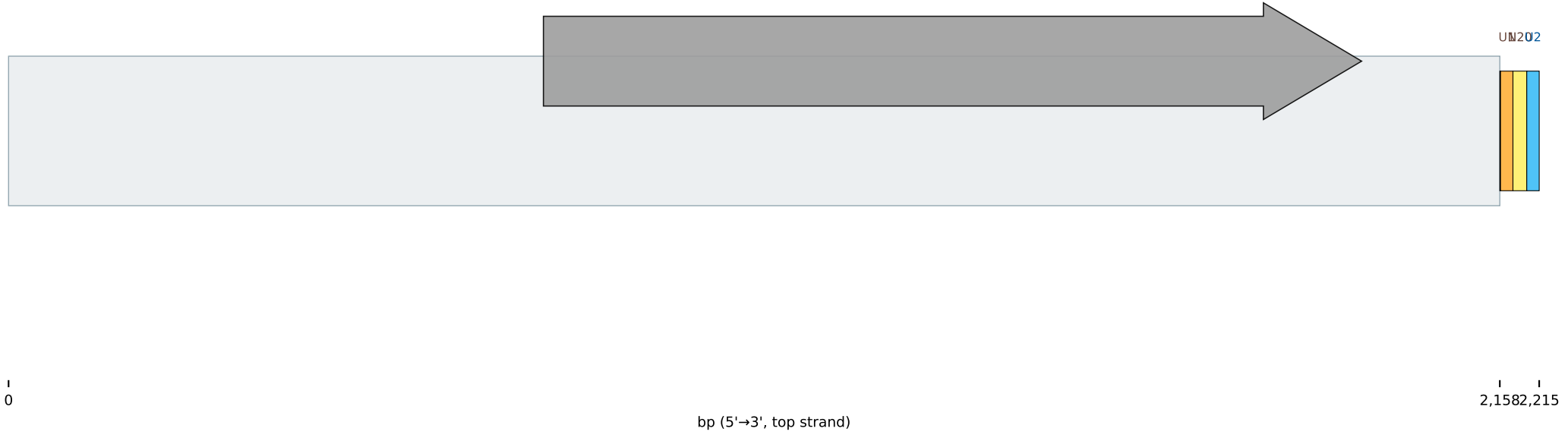
annotations: 0 CDS + 0 misc (pyrodigal fallback)

```
Tier / category:     Tier 3: Cross-phylum BHR
Rep family:          REP_ORF2-G2P__RCR
Taxonomy:            Gammaproteobacteria / Enterobacteriaceae
GC content:          0.5543323863636364
Barcode (N20):       TACAGGTTTGCGATAGCTGG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,221 bp (origin 3,164 + cassette 57)
Flags:               none
```

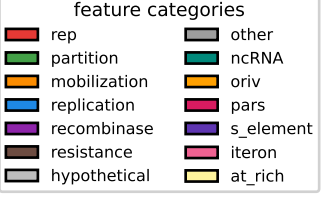


1883660858 | Polaromonas sp. W11N | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NZ_MG869624.1))

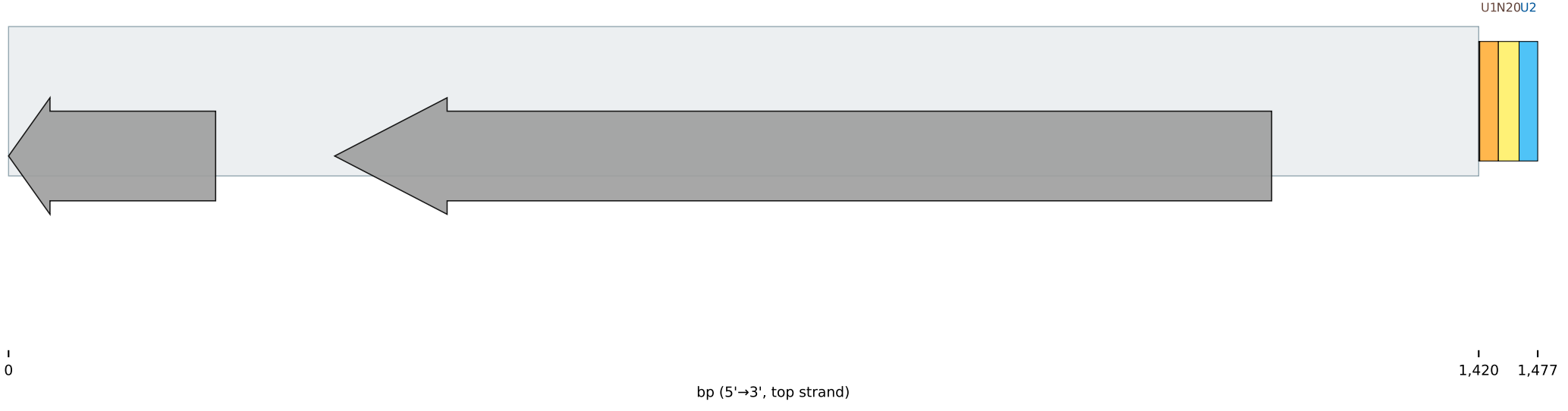



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep_1__RCR
Taxonomy:            Betaproteobacteria / Comamonadaceae
GC content:          0.5775606867969213
Barcode (N20):       CGTTACTTCTGGTTAGTGTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,215 bp (origin 2,158 + cassette 57)
Flags:               none
```

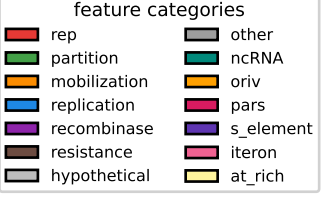


1908110475 | Defluviicoccus vanus | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_CP053924.1))



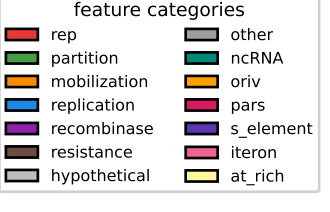

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RepABC_RepC
Taxonomy:            Alphaproteobacteria / Rhodospirillaceae
GC content:          0.6431843891402715
Barcode (N20):       TGAACGCATCCTGATTCATG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,477 bp (origin 1,420 + cassette 57)
Flags:               none
```



1918150558 | Rhodococcus erythropolis | diversity_PLSDB | twist_synthesis

annotations: 0 CDS + 0 misc (pyrodigal fallback)

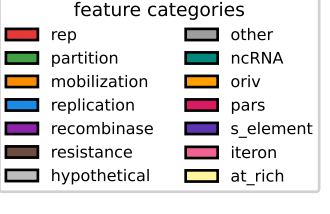
```
Tier / category:     Tier 3: Cross-phylum BHR
Rep family:          REP_ORF2-G2P__RCR
Taxonomy:            Actinomycetes / Nocardiaceae
GC content:          0.5942321822189567
Barcode (N20):       GATATTATGCCGTTAGCTGA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,566 bp (origin 2,509 + cassette 57)
Flags:               none
```



1928270379 | Rothia terrae | diversity_PLSDB | twist_synthesis

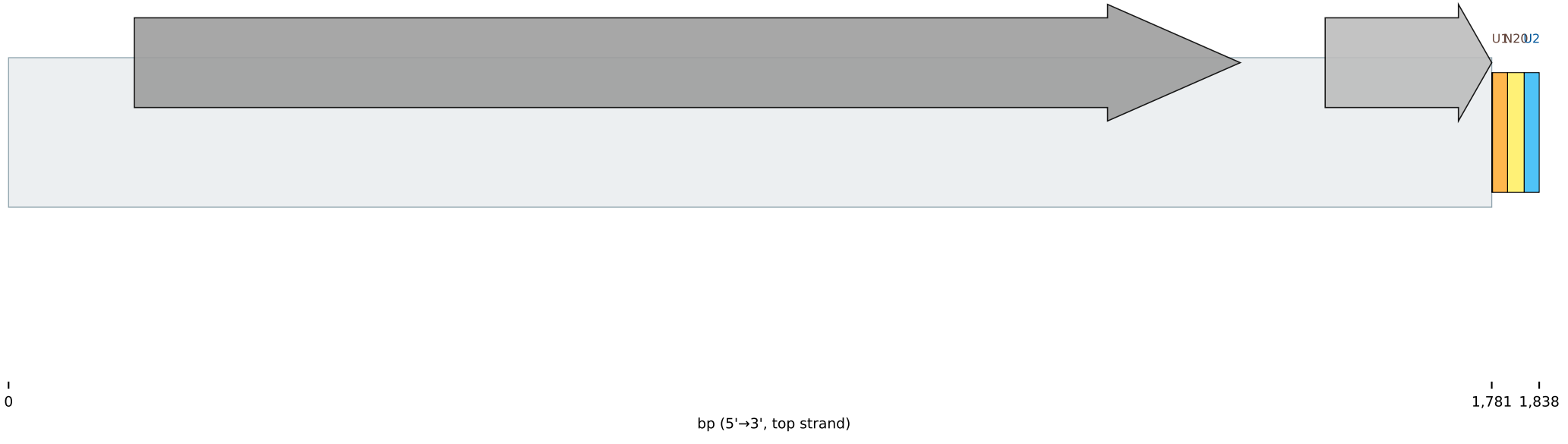
annotations: 0 CDS + 0 misc (pyrodigal fallback)

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          PriCT_1
Taxonomy:            Actinomycetes / Micrococcaceae
GC content:          0.4996031746031746
Barcode (N20):       TACAGGTCTTAAGTGACATG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,061 bp (origin 2,004 + cassette 57)
Flags:               none
```

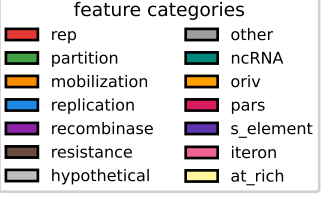


1944259351 | Aeromonas sp. | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_MK907532.1))

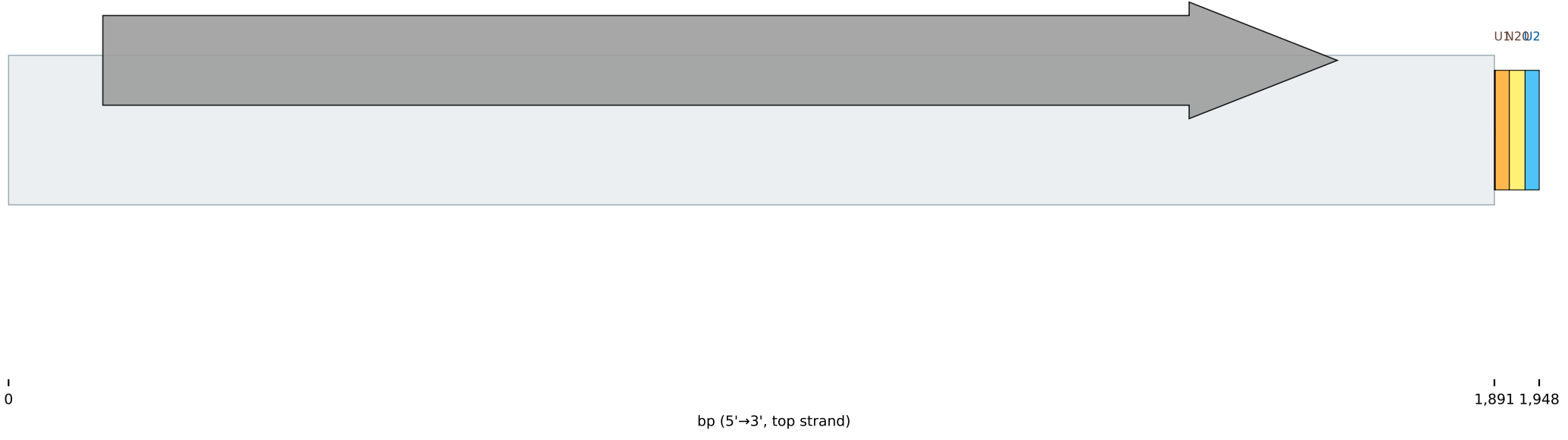



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep_1__RCR
Taxonomy:            Gammaproteobacteria / Aeromonadaceae
GC content:          0.5867949278530826
Barcode (N20):       TATAAACATGATCCGCCACA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,838 bp (origin 1,781 + cassette 57)
Flags:               none
```

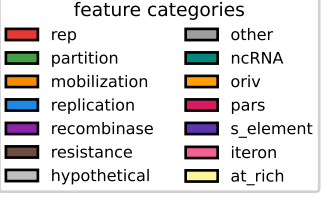


1977563761 | Aeromonas sp. | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NZ_MK962686.1))



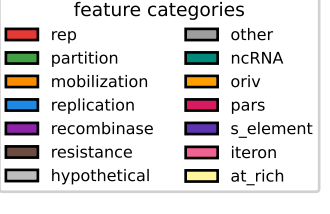

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep_1__RCR
Taxonomy:            Gammaproteobacteria / Aeromonadaceae
GC content:          0.597103357472021
Barcode (N20):       GGCAAGACTTTGAGGTGTAA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,948 bp (origin 1,891 + cassette 57)
Flags:               none
```



1977568769 | Aeromonas sp. | diversity_PLSDB | twist_synthesis

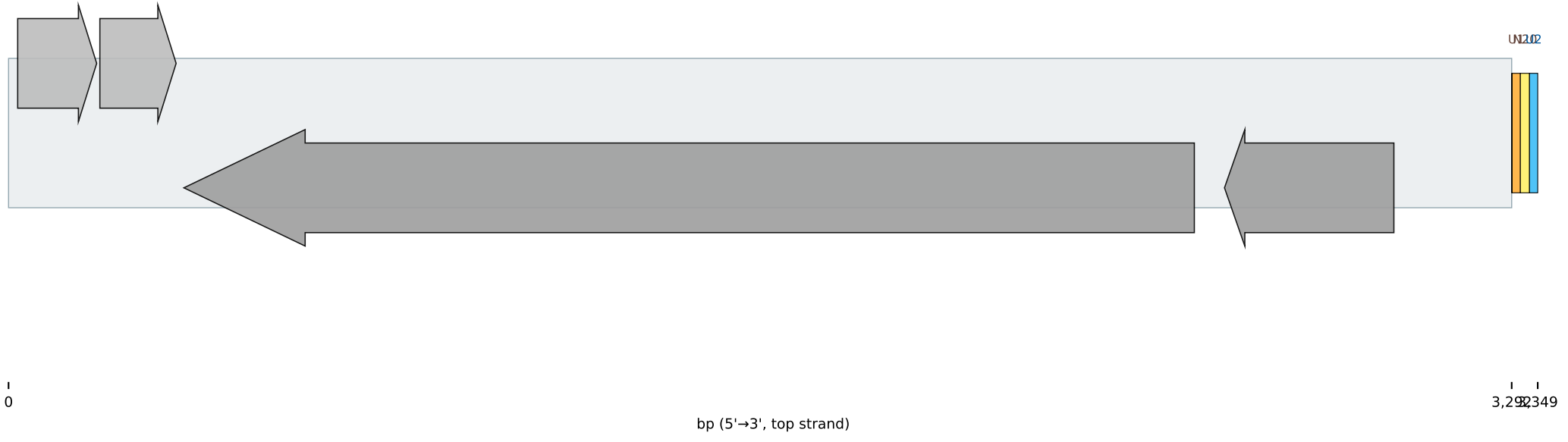
annotations: 0 CDS + 0 misc (pyrodigal fallback)

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RepL__RCR
Taxonomy:            Gammaproteobacteria / Aeromonadaceae
GC content:          0.5795090715048026
Barcode (N20):       AGGGCTGTAGTCAGCTCCTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,511 bp (origin 2,454 + cassette 57)
Flags:               none
```



2163906054 | Halobacterium sp. BOL4-2 | diversity_PLSDB | twist_synthesis

annotations: 4 CDS + 0 misc (.gb (NZ_CP070335.2))




```
Tier / category:     Tier 4: Phylum-exclusive
Rep family:          RNA_helicase__RCR
Taxonomy:            Halobacteria / Halobacteriaceae
GC content:          0.5914577375251533
Barcode (N20):       GTAGAGCCTGAAGGATCCTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,349 bp (origin 3,292 + cassette 57)
Flags:               none
```

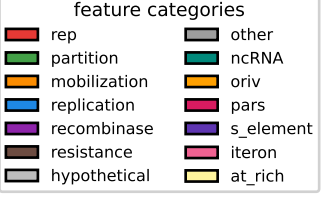
feature categories

rep	other
partition	ncRNA
mobilization	oriv
replication	pars
recombinase	s_element
resistance	iteron
hypothetical	at_rich

2184550974 | Aeromonas caviae | diversity_PLSDB | twist_synthesis

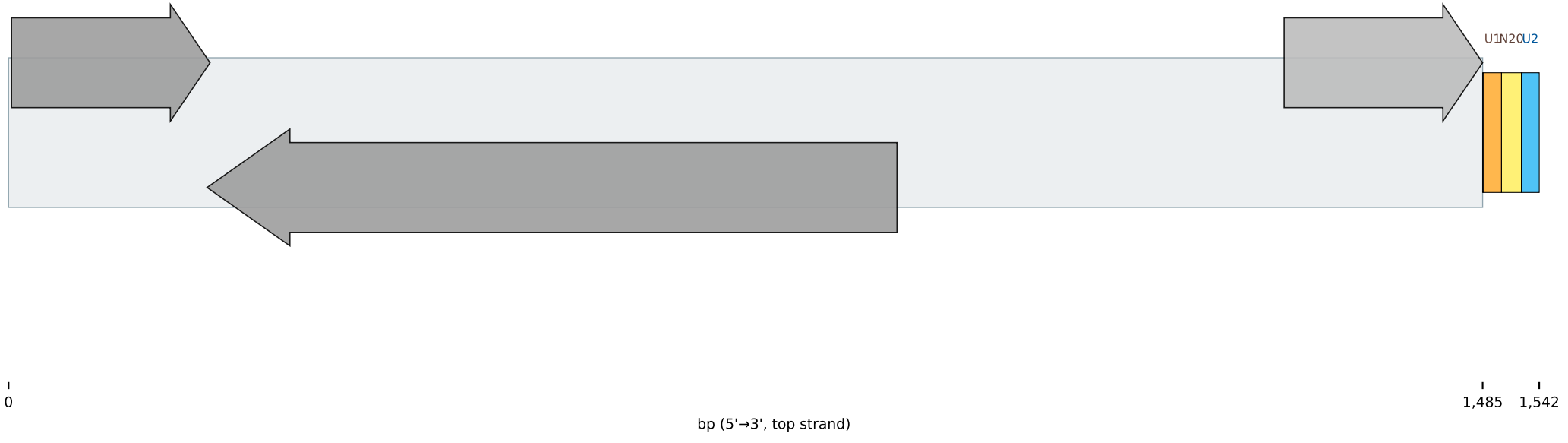
annotations: 4 CDS + 0 misc (.gb (NZ_CP091178.1))

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RepL__RCR
Taxonomy:            Gammaproteobacteria / Aeromonadaceae
GC content:          0.5833537713763349
Barcode (N20):       CATAACTATGTAGCCCGAGA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,992 bp (origin 1,935 + cassette 57)
Flags:               none
```

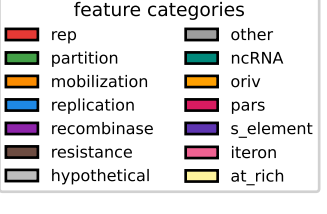


2302881874 | Sulfitobacter pontiacus | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_CP081118.1))

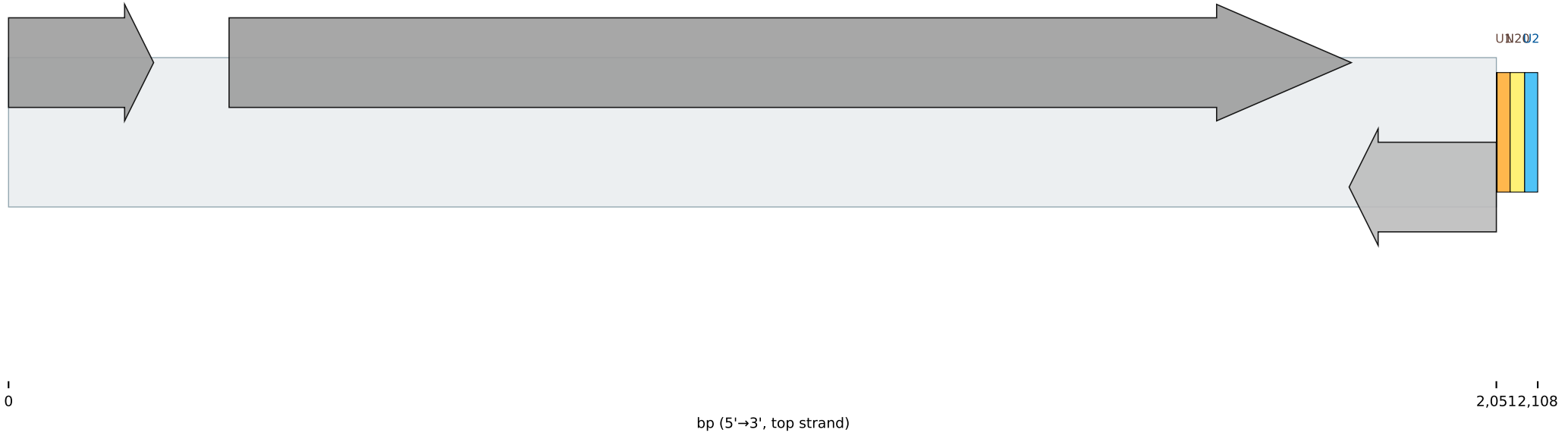



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          DnaA_N
Taxonomy:            Alphaproteobacteria / Roseobacteraceae
GC content:          0.5931279503806829
Barcode (N20):       TCATCTGCGCGATAGTAGTG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,542 bp (origin 1,485 + cassette 57)
Flags:               none
```

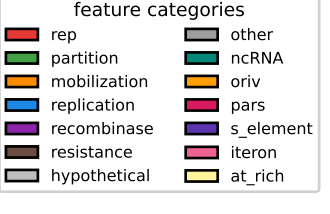


2311204984 | Rhodococcus erythropolis | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_AP018736.1))



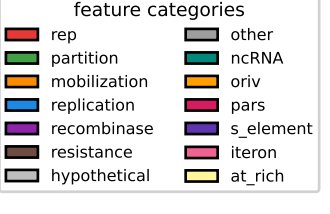

```
Tier / category:     Tier 3: Cross-phylum BHR
Rep family:          REP_ORF2-G2P__RCR
Taxonomy:            Actinomycetes / Nocardiaceae
GC content:          0.5942321822189567
Barcode (N20):       GTGTCGTACTCTGGCCTGAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,108 bp (origin 2,051 + cassette 57)
Flags:               none
```



2437001263 | Edwardsiella piscicida | diversity_PLSDB | twist_synthesis

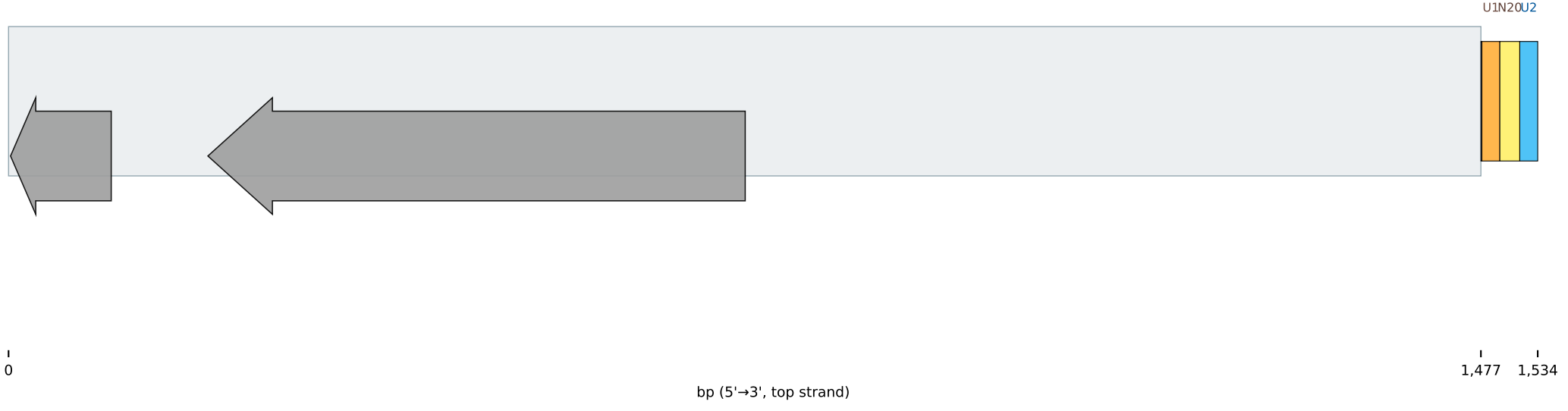
annotations: 2 CDS + 0 misc (.gb (CP104739.1))

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RepL__RCR
Taxonomy:            Gammaproteobacteria / Hafniaceae
GC content:          0.5750602673679597
Barcode (N20):       TCTGGACGAACTAACACTAG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,130 bp (origin 2,073 + cassette 57)
Flags:               none
```

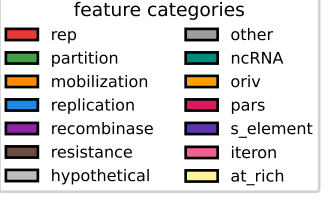


2442125848 | Paracoccus fistulariae | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NZ_CP067139.1))

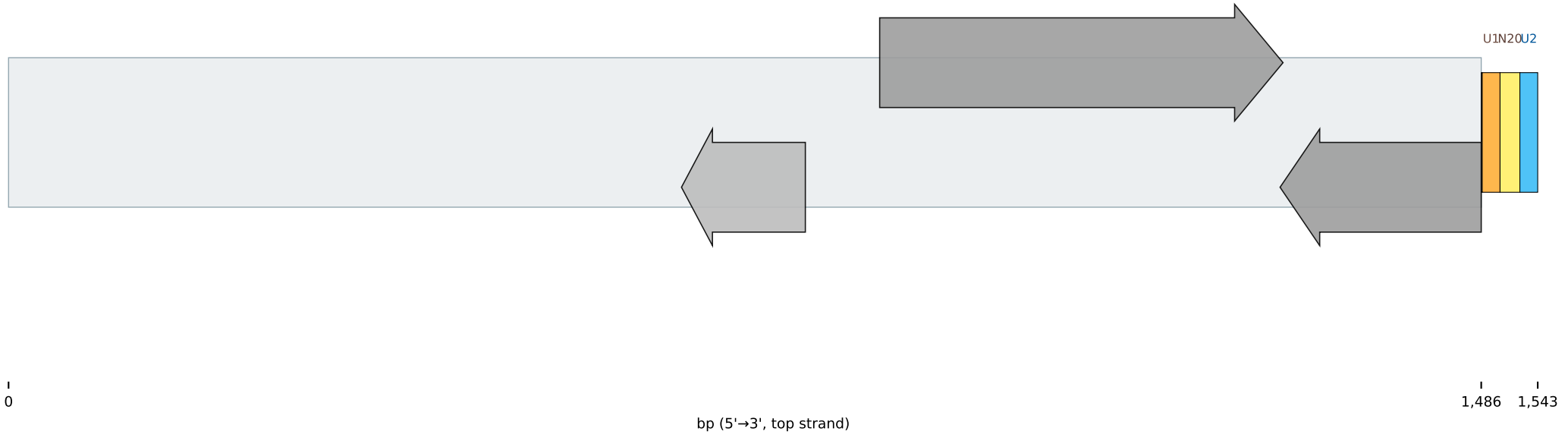



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RepL__RCR
Taxonomy:            Alphaproteobacteria / Paracoccaceae
GC content:          0.5887720690339219
Barcode (N20):       AGCGGTTCGGCATAACAACG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,534 bp (origin 1,477 + cassette 57)
Flags:               none
```

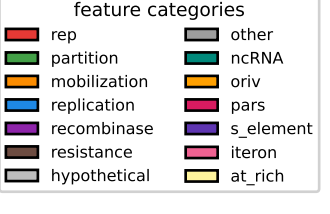


2473170302 | Sulfitobacter pontiacus | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (AP027374.1))

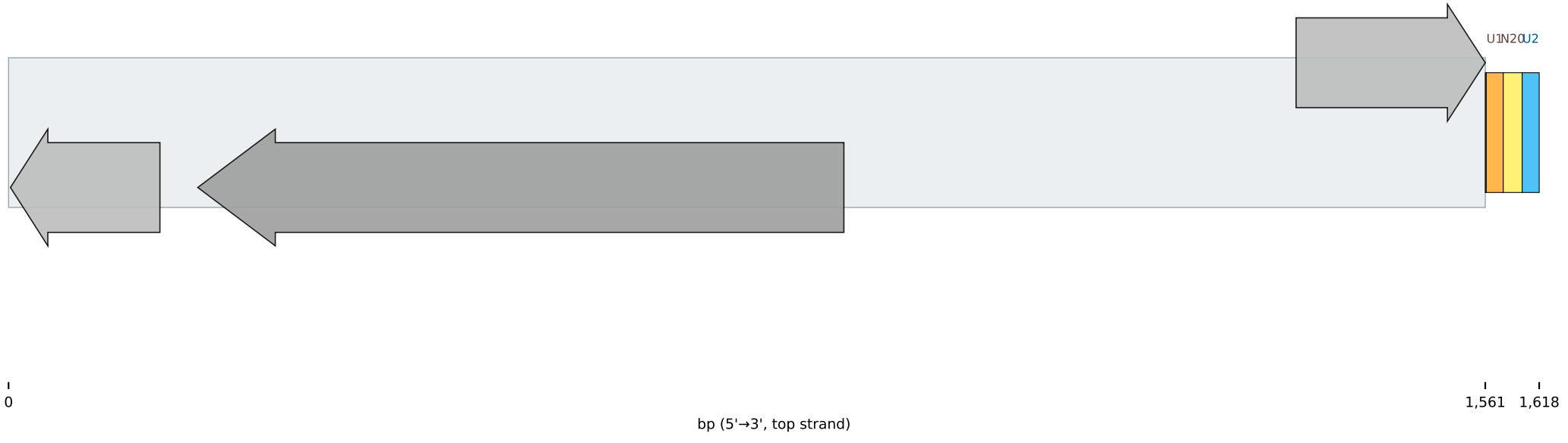



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          DnaA_N
Taxonomy:            Alphaproteobacteria / Roseobacteraceae
GC content:          0.5921053529158964
Barcode (N20):       ATATATTTGCTACCTGAGCC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,543 bp (origin 1,486 + cassette 57)
Flags:               none
```

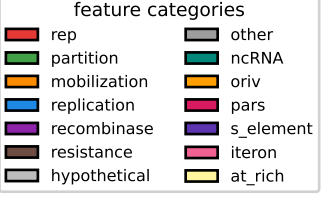


2520865930 | Marivivens sp. LCG002 | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NZ_CP127166.1))

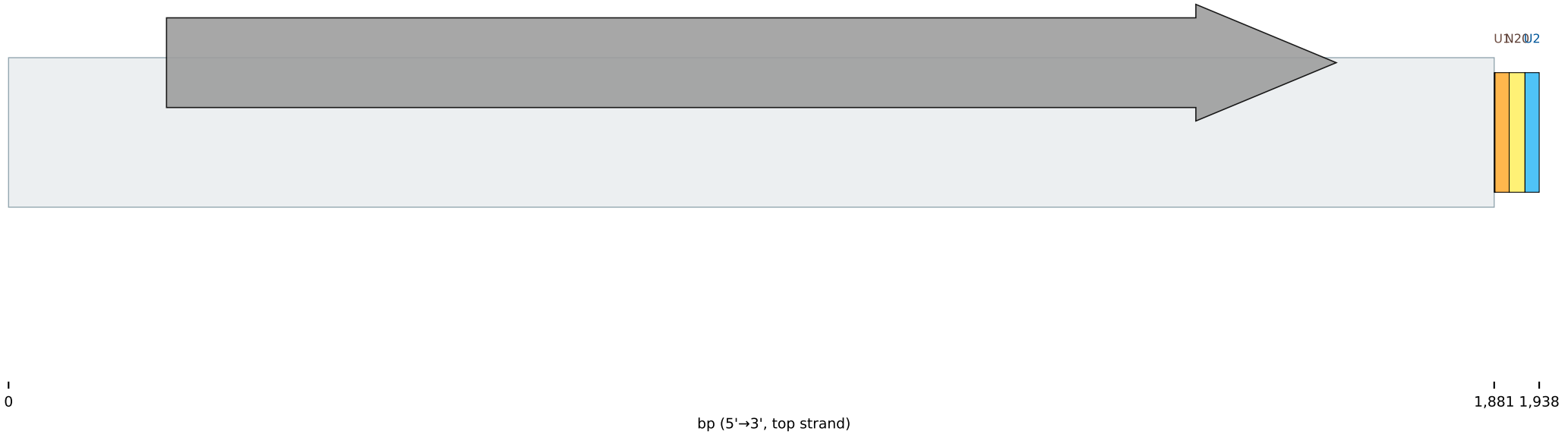



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          DnaA_N
Taxonomy:            Alphaproteobacteria / Paracoccaceae
GC content:          0.5870509486025112
Barcode (N20):       AACCATCAACGCTAGAGTCT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,618 bp (origin 1,561 + cassette 57)
Flags:               none
```



2548995762 | Lactiplantibacillus pentosus | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NZ_CP129659.1))



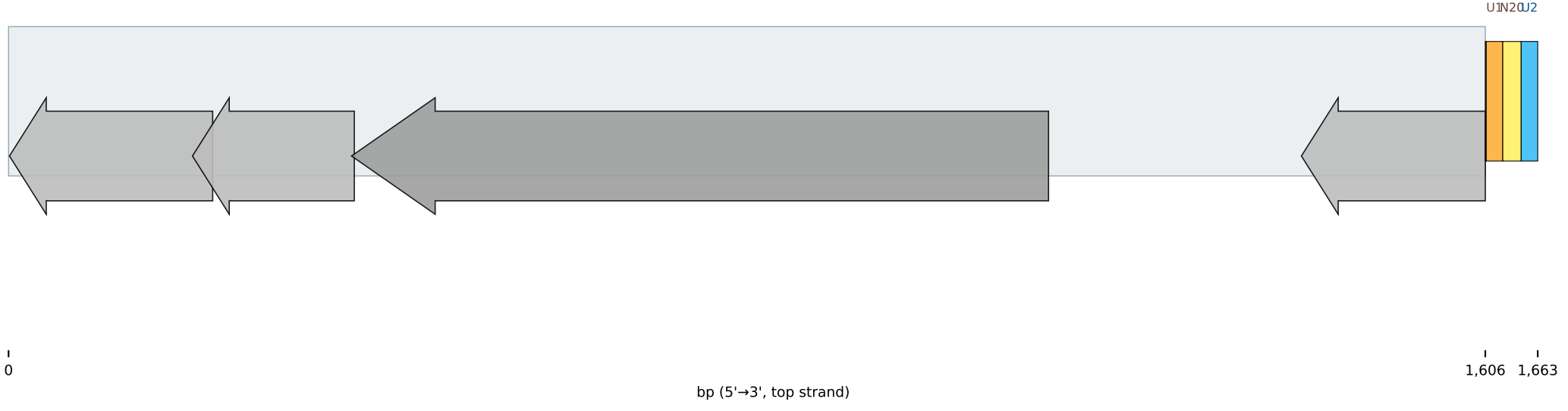

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          PriCT_1
Taxonomy:            Bacilli / Lactobacillaceae
GC content:          0.5059665871121718
Barcode (N20):       TTGTCCTCGTAGCGGTACCC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,938 bp (origin 1,881 + cassette 57)
Flags:               none
```

feature categories

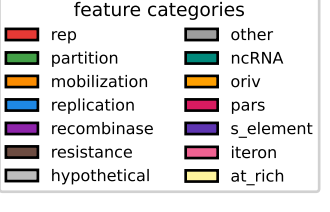
- rep
- partition
- mobilization
- replication
- recombinase
- resistance
- hypothetical
- other
- ncRNA
- oriv
- pars
- s_element
- iteron
- at_rich

2581911996 | Subdoligranulum variabile | diversity_PLSDB | twist_synthesis

annotations: 4 CDS + 0 misc (.gb (OY725836.1))

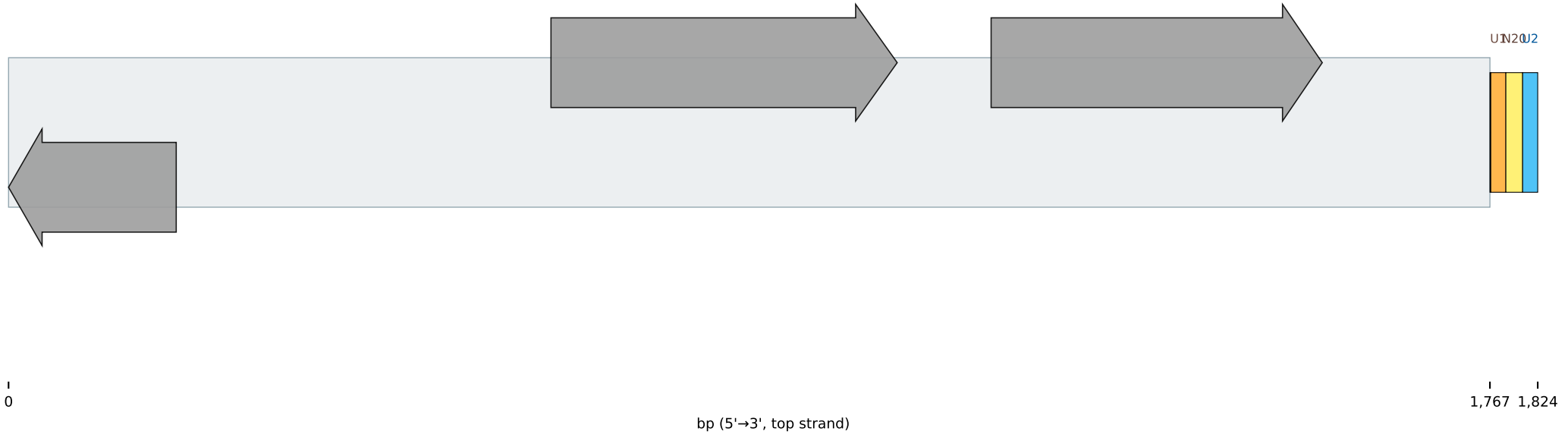



```
Tier / category:     Tier 3: Cross-phylum BHR
Rep family:          REP_ORF2-G2P__RCR
Taxonomy:            Clostridia / Oscillospiraceae
GC content:          0.5784101273683462
Barcode (N20):       TGCATCGACATGAAGTCCAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,663 bp (origin 1,606 + cassette 57)
Flags:               none
```

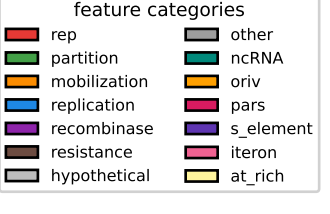


2581912037 | Subdoligranulum variabile | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (OY725877.1))



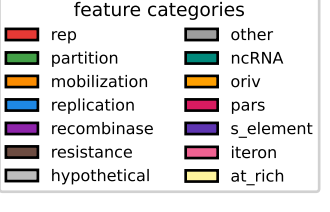

```
Tier / category:     Tier 4: Phylum-exclusive
Rep family:          RepA_N
Taxonomy:            Clostridia / Oscillospiraceae
GC content:          0.5883450127905975
Barcode (N20):       CTGAATCCTACCTCCAAGCG
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    1,824 bp (origin 1,767 + cassette 57)
Flags:               none
```



2581912041 | Subdoligranulum variabile | diversity_PLSDB | twist_synthesis

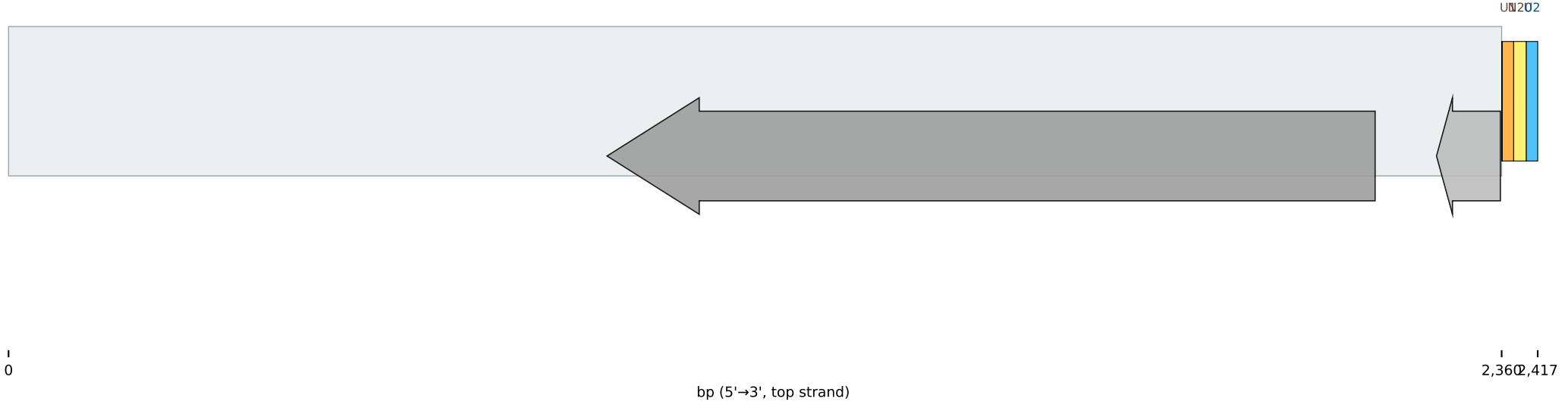
annotations: 0 CDS + 0 misc (pyrodigal fallback)

```
Tier / category:     Tier 4: Phylum-exclusive
Rep family:          RepA_N
Taxonomy:            Clostridia / Oscillospiraceae
GC content:          0.5492251758254858
Barcode (N20):       CGACAGCATTCATGCAAGTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    6,057 bp (origin 6,000 + cassette 57)
Flags:               OVERSIZE >5 kb (Ian to re-review)
```

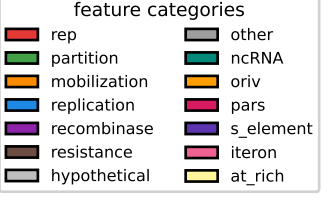


2639748091 | Sinorhizobium meliloti | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (CP140962.1))

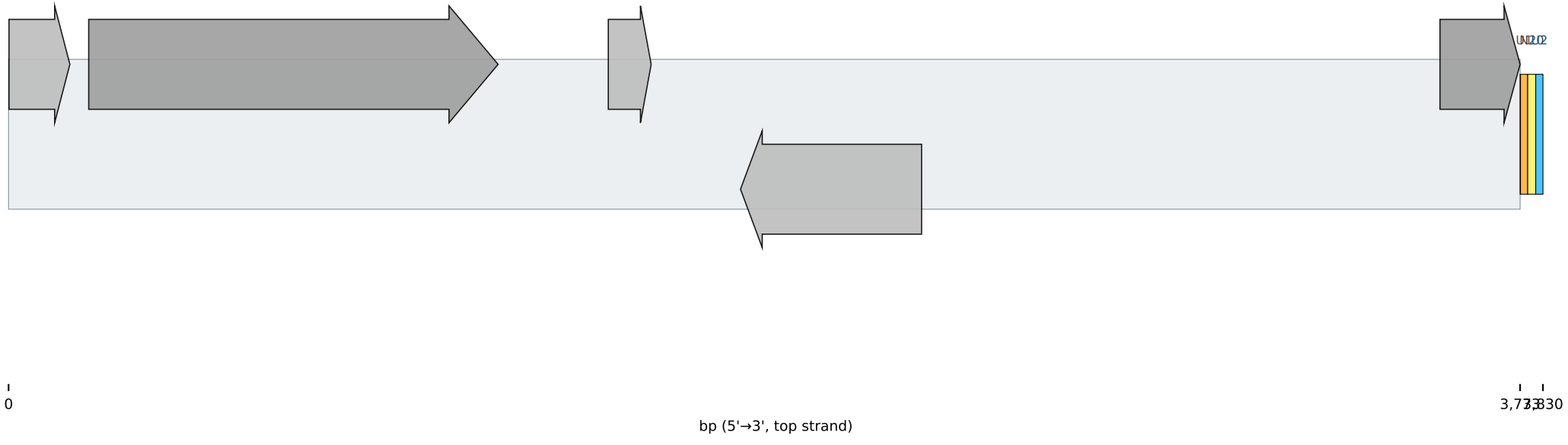



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          PRK13824.1
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.5931221832861178
Barcode (N20):       AAACCGCTCGTTTCCACGCT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,417 bp (origin 2,360 + cassette 57)
Flags:               none
```

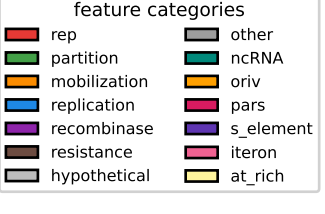


2732292344 | Paraburkholderia sp. PREW-6R | diversity_PLSDB | twist_synthesis

annotations: 5 CDS + 0 misc (.gb (NZ_CP155076.1))

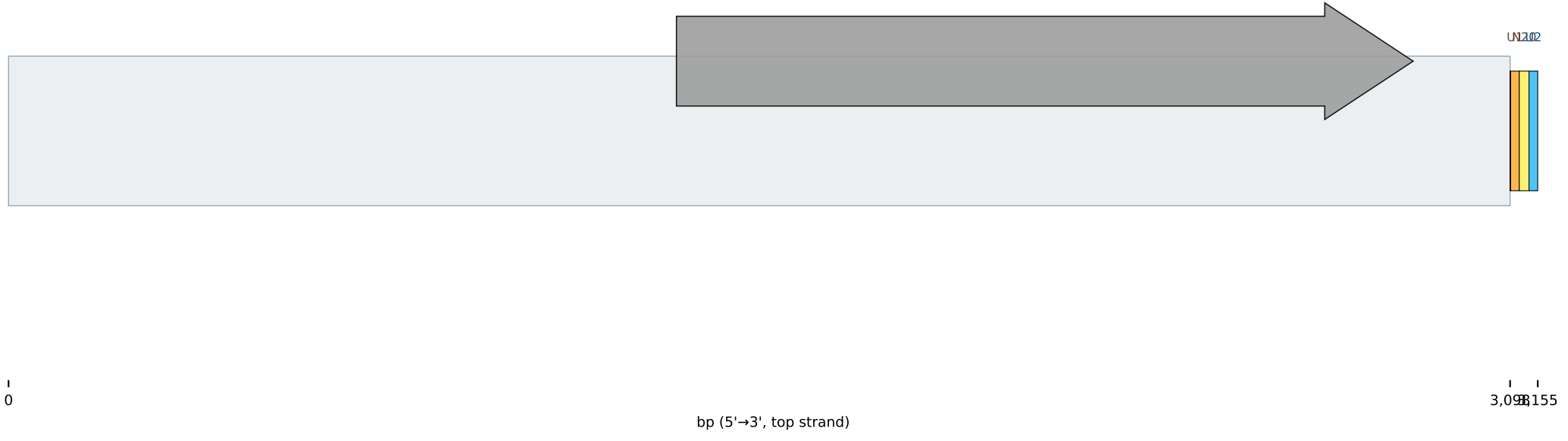



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          RPA
Taxonomy:            Betaproteobacteria / Burkholderiaceae
GC content:          0.5925969010283374
Barcode (N20):       GACAGTGTATGCGGAAACTT
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,830 bp (origin 3,773 + cassette 57)
Flags:               none
```

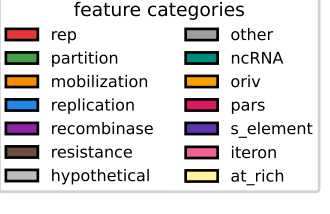


32455369 | Sinorhizobium meliloti | diversity_PLSDB | twist_synthesis

annotations: 1 CDS + 0 misc (.gb (NC_004965.1))

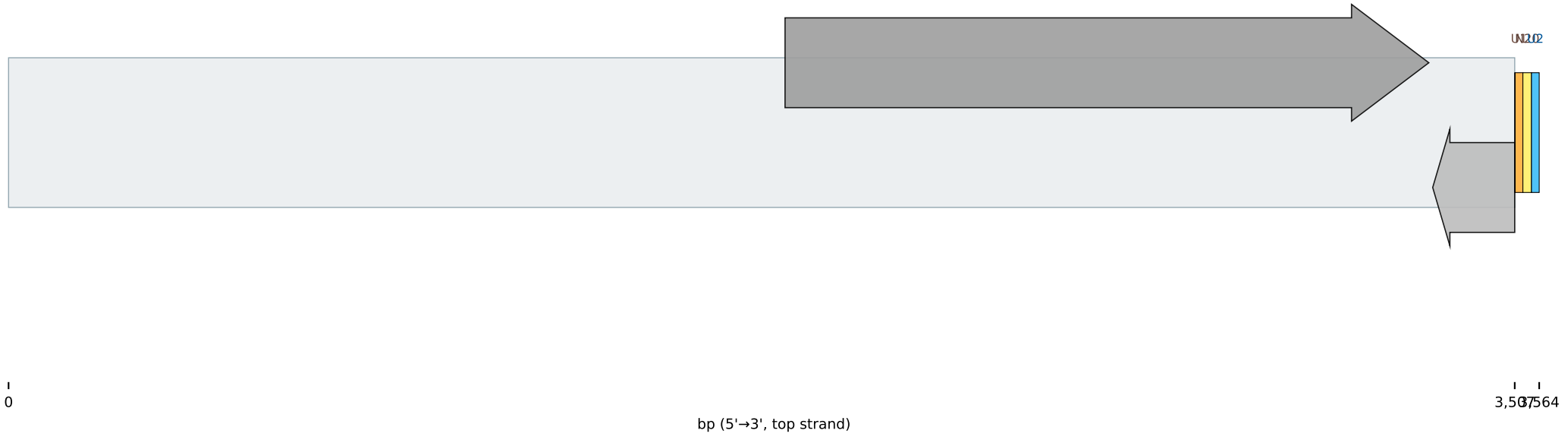



```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep_1__RCR
Taxonomy:            Alphaproteobacteria / Rhizobiaceae
GC content:          0.6163338879645036
Barcode (N20):       CTTGGCCTTGATACGACGAC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,155 bp (origin 3,098 + cassette 57)
Flags:               none
```



357941606 | Burkholderia sp. YI23 | diversity_PLSDB | twist_synthesis

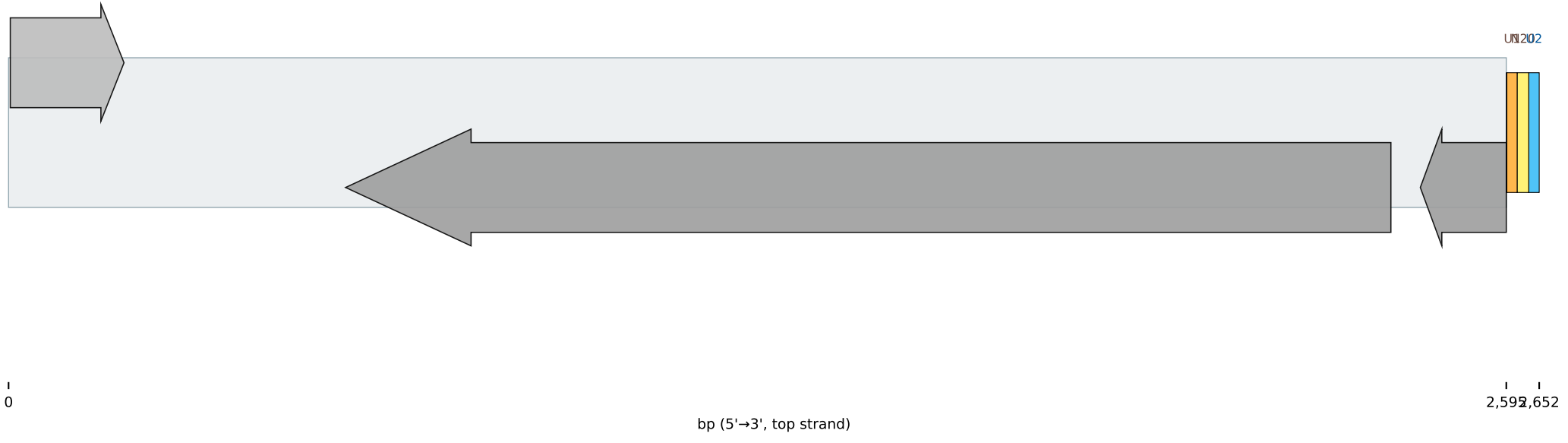
annotations: 2 CDS + 0 misc (.gb (CP003091.1))



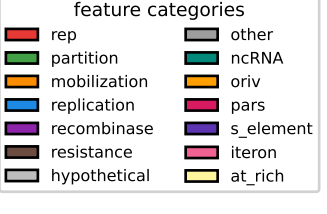

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep3_N
Taxonomy:            Betaproteobacteria / Burkholderiaceae
GC content:          0.5873329534641543
Barcode (N20):       CTTCATTGAGATGAGTTCGC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    3,564 bp (origin 3,507 + cassette 57)
Flags:               none
```

56708791 | Ruegeria pomeroyi | diversity_PLSDB | twist_synthesis

annotations: 3 CDS + 0 misc (.gb (NC_006569.1))

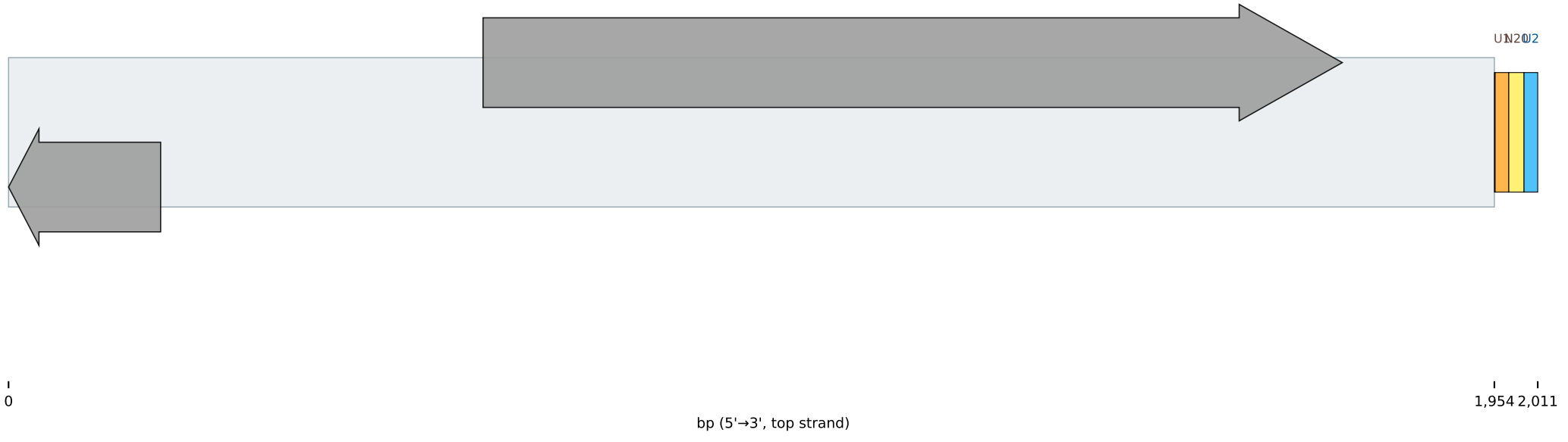


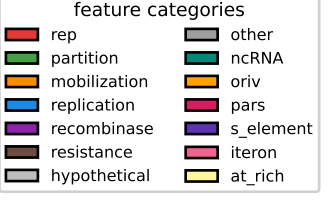

```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          PriCT_1
Taxonomy:            Alphaproteobacteria / Roseobacteraceae
GC content:          0.6278826145061848
Barcode (N20):       CCTGGAAATGTTACCGCTTC
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,652 bp (origin 2,595 + cassette 57)
Flags:               none
```



84798614 | Paramagnetospirillum magneticum | diversity_PLSDB | twist_synthesis

annotations: 2 CDS + 0 misc (.gb (NC_007706.1))




```
Tier / category:     Tier 2: Alphaproteobacteria
Rep family:          Rep_1__RCR
Taxonomy:            Alphaproteobacteria / Magnetospirillaceae
GC content:          0.5918203688853247
Barcode (N20):       CCAACACTAGAGGCAGATCA
Cassette (3' tail):  A | GATGTCCACGAGGTCTCT | N20 | CGTACGCTGCAGGTCGAC
Ordered fragment:    2,011 bp (origin 1,954 + cassette 57)
Flags:               none
```